Natural Variability of Climate

Global Climate Change – Week 2

Dr Thomas Bannan

Thomas.Bannan@Manchester.ac.uk

Today's Lecture

Part 1: Natural Variability (60 mins)

Part 2: Break (10 mins)

Part 3: Q/A (10 mins)

Part 4: Workshop (40 mins)



Global Climate Change

The University of Manchester

Week 1 – Introduction to climate change and its impacts (5th Feb)

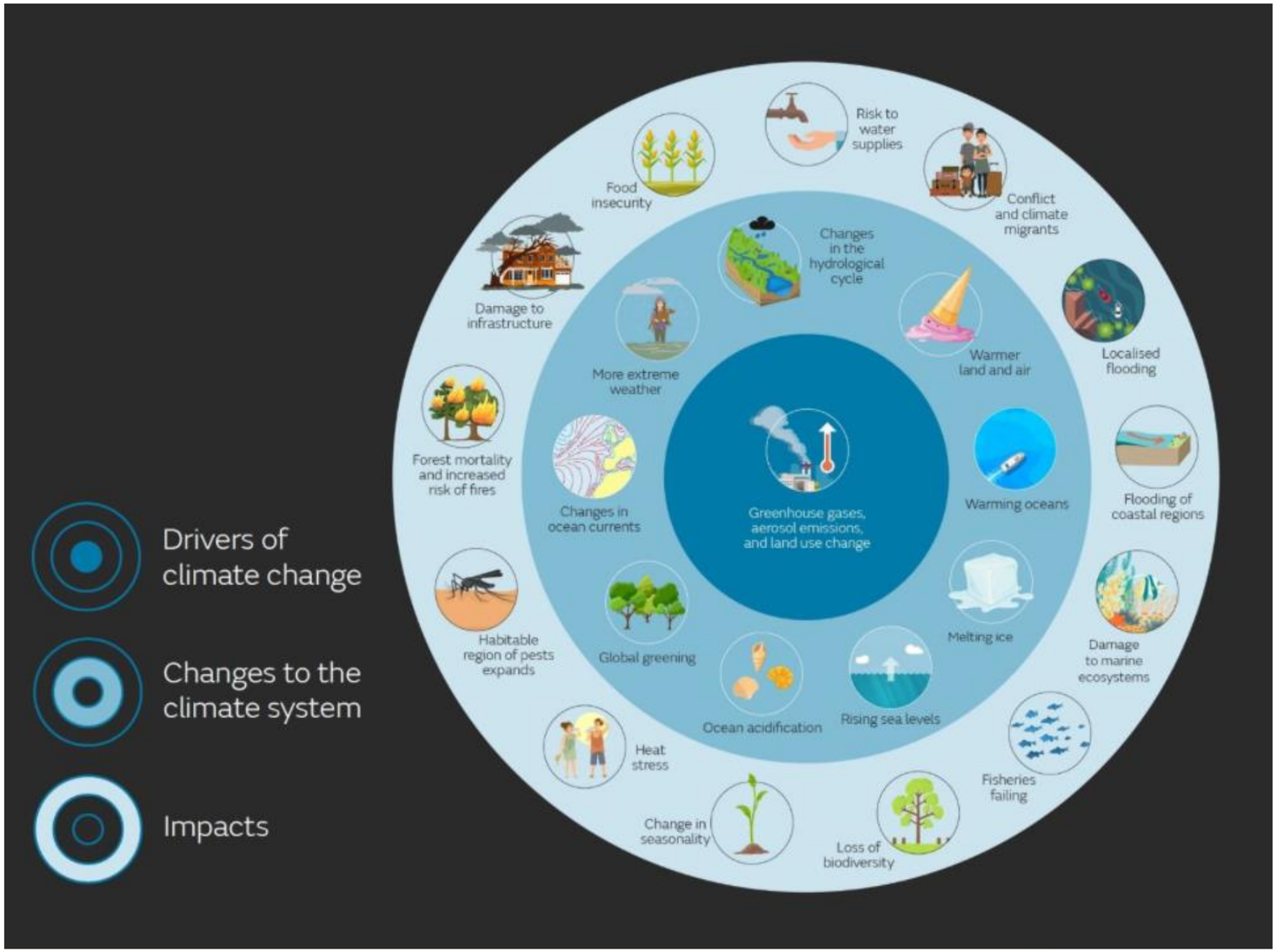
Week 2 – Natural Climate Variability (12th Feb)

Week 3 – Radiation Law/Greenhouse Effect/Radiative forcing/feedbacks (19th Feb)
Week 4 – Anthropogenic impacts (26th Feb)
Week 5 – Observational evidence and uncertainties on climate change (5th March)
Week 6 – Mid Unit Test (10th March)
Week 7 – Climate Models (19th March)
Week 8 – Climate Change Adaptation (26th March)
Week 9 – Effects of Climate Change: Case Studies (23rd April)
Week 10 – Climate Change Mitigation (30th April)
Week 11 – Revision Lecture (7th May)

Comments/Questions

Last Week: Drivers, Changes and Impacts

The University of Manchester

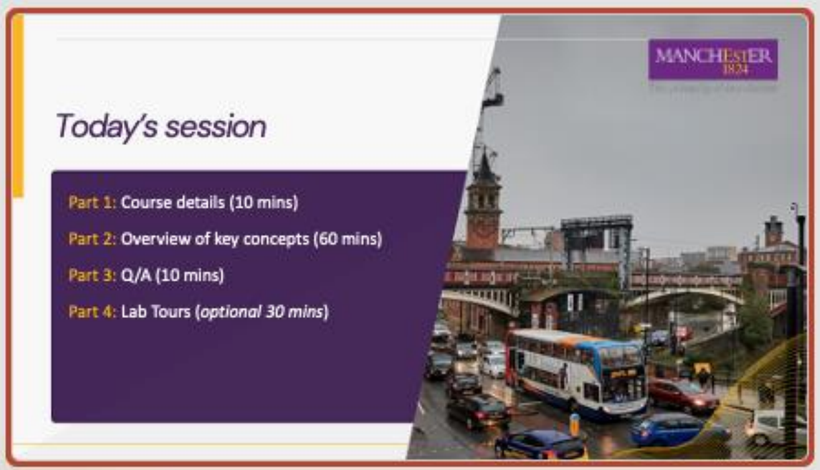


1. **Human Drivers:** Fossil fuel use, land-use change, and industry are the main sources of greenhouse gas emissions.
2. **Key Gases:** CO_2 is the dominant greenhouse gas, with methane and nitrous oxide also contributing significantly.
3. **Other Forcers:** Short-lived climate forcers, including aerosols and black carbon, have complex climate effects.
4. **Rising Levels:** Greenhouse gas emissions and concentrations have increased sharply since pre-industrial times.
5. **Radiative Forcing:** Human activities drive positive radiative forcing, partly offset by cooling from aerosols.
6. **Future Impacts:** Without mitigation, warming beyond 1.5–2°C is expected, with regionally variable but widespread impacts.

Lecture Structure

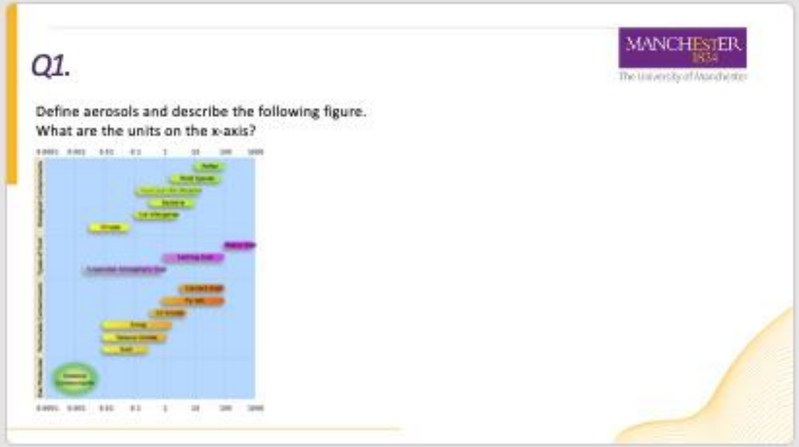
The University of Manchester

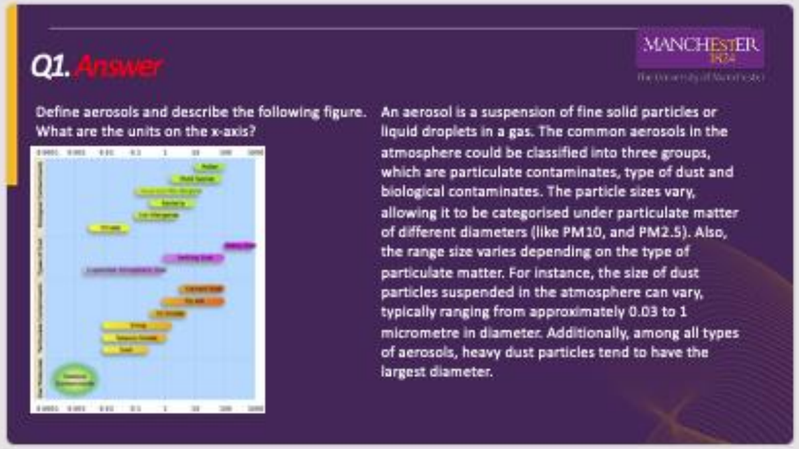
1 Hour Lecture





1 Hour Workshop







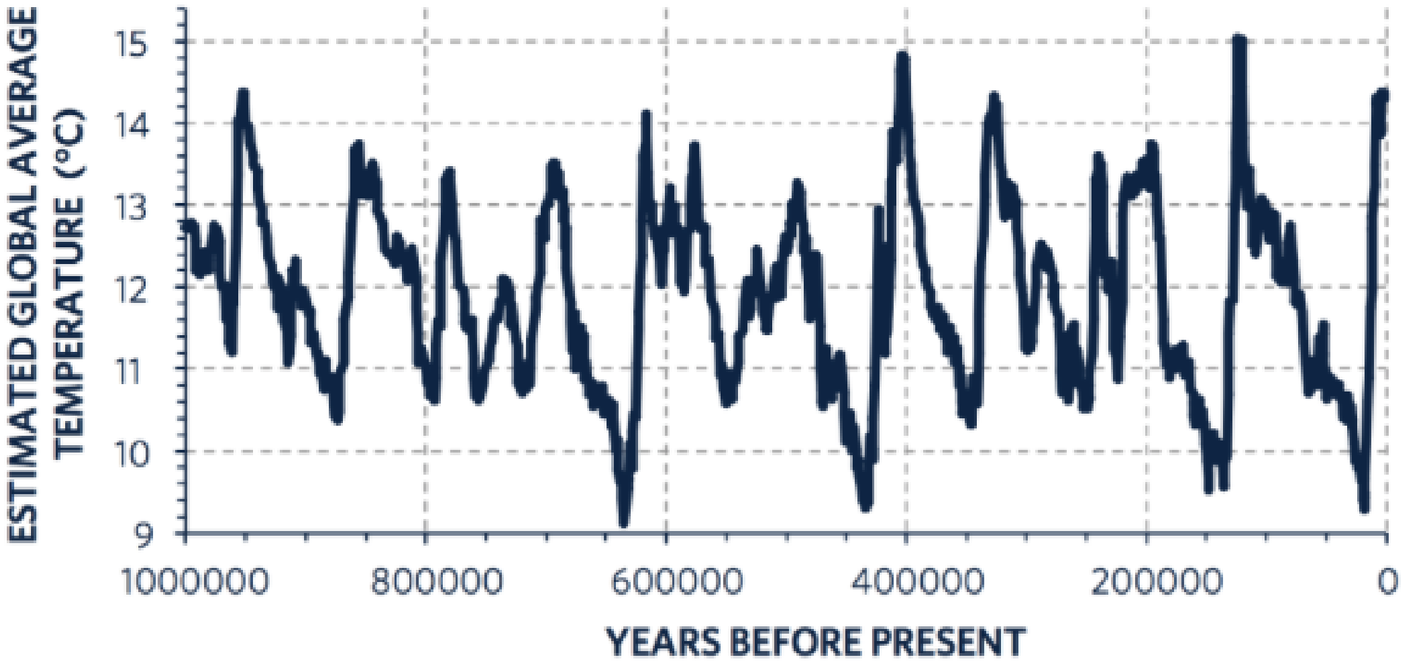
Intended Learning Outcomes

By the end of this week, you will be able to:

- **Say what natural climate variability is** and **tell the difference** between natural changes and human-driven climate change.
- **Explain the main natural causes** of climate variation, including Earth's orbit, the Sun, volcanoes, and the oceans.
- **Describe how Milankovitch cycles work** (changes in Earth's orbit and tilt) and how they can lead to ice ages and warmer periods.
- **Explain what solar changes and volcanic eruptions do to climate**, and how long their effects usually last.
- **Describe ENSO (El Niño/La Niña)** and **explain how it affects weather and climate around the world.**
- **Use evidence from past climates** (like ice cores and other proxies) to **interpret natural climate changes** and connect them to what we see today *(developed further in week 5)*

Our key question today: *what drives natural variability in climate?*





Variations in Energy Inputs

Two types of changes

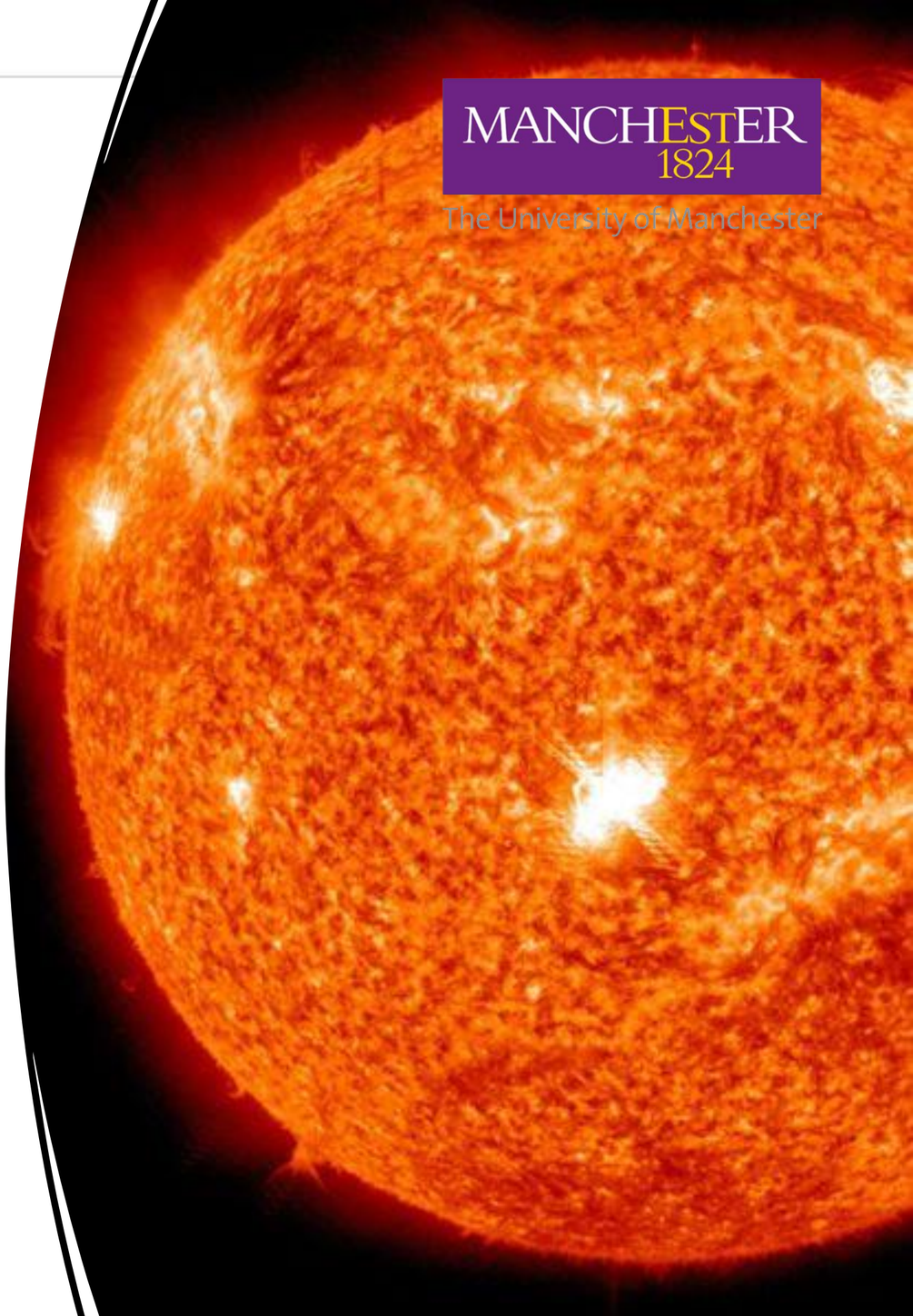
- Alterations to the Sun-Earth system
- Alterations in the Solar output

Sun-Earth system changes lead to:

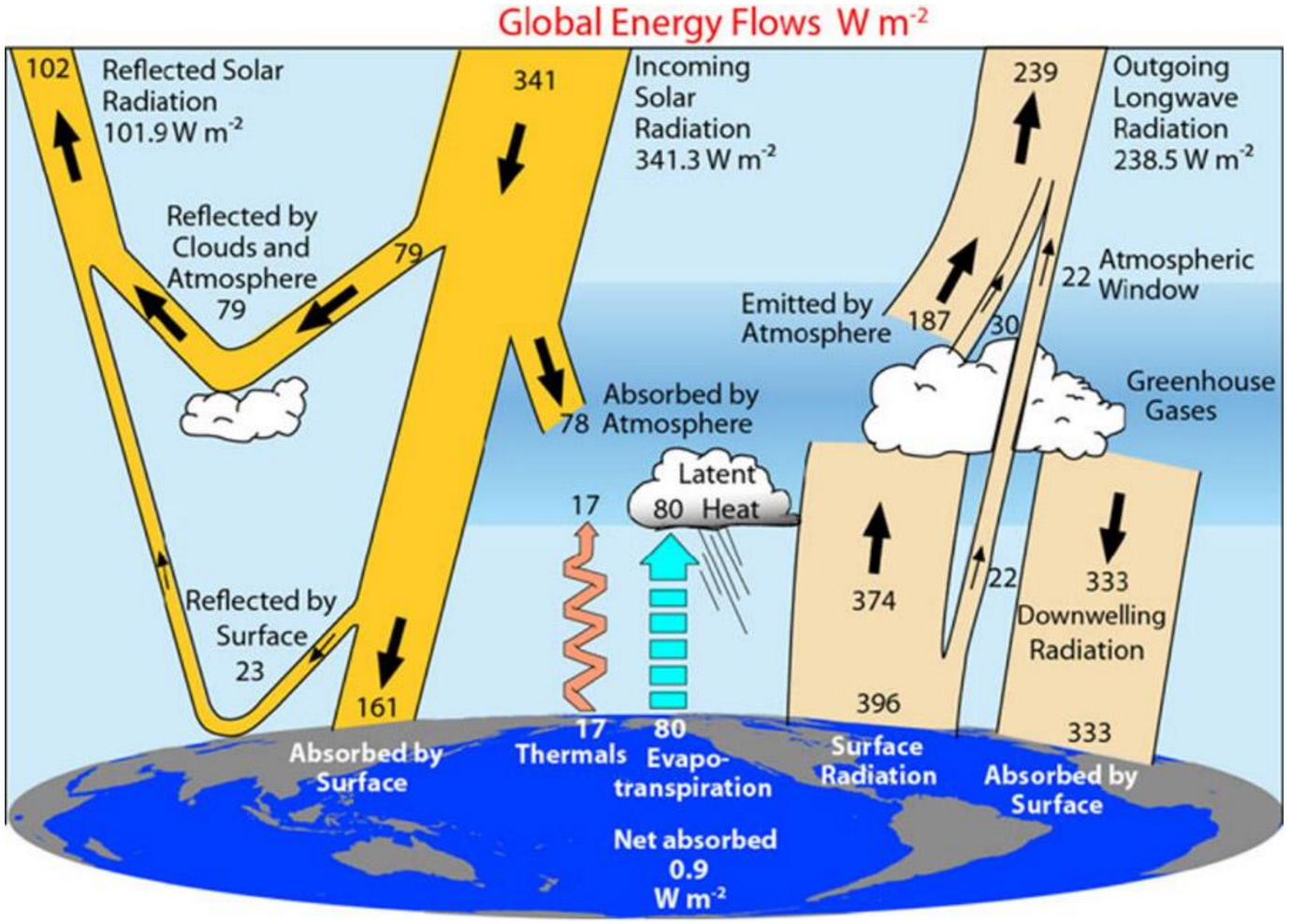
- Seasonal variations, prolonged periods of cooler or warmer climates, variations in daylight hours

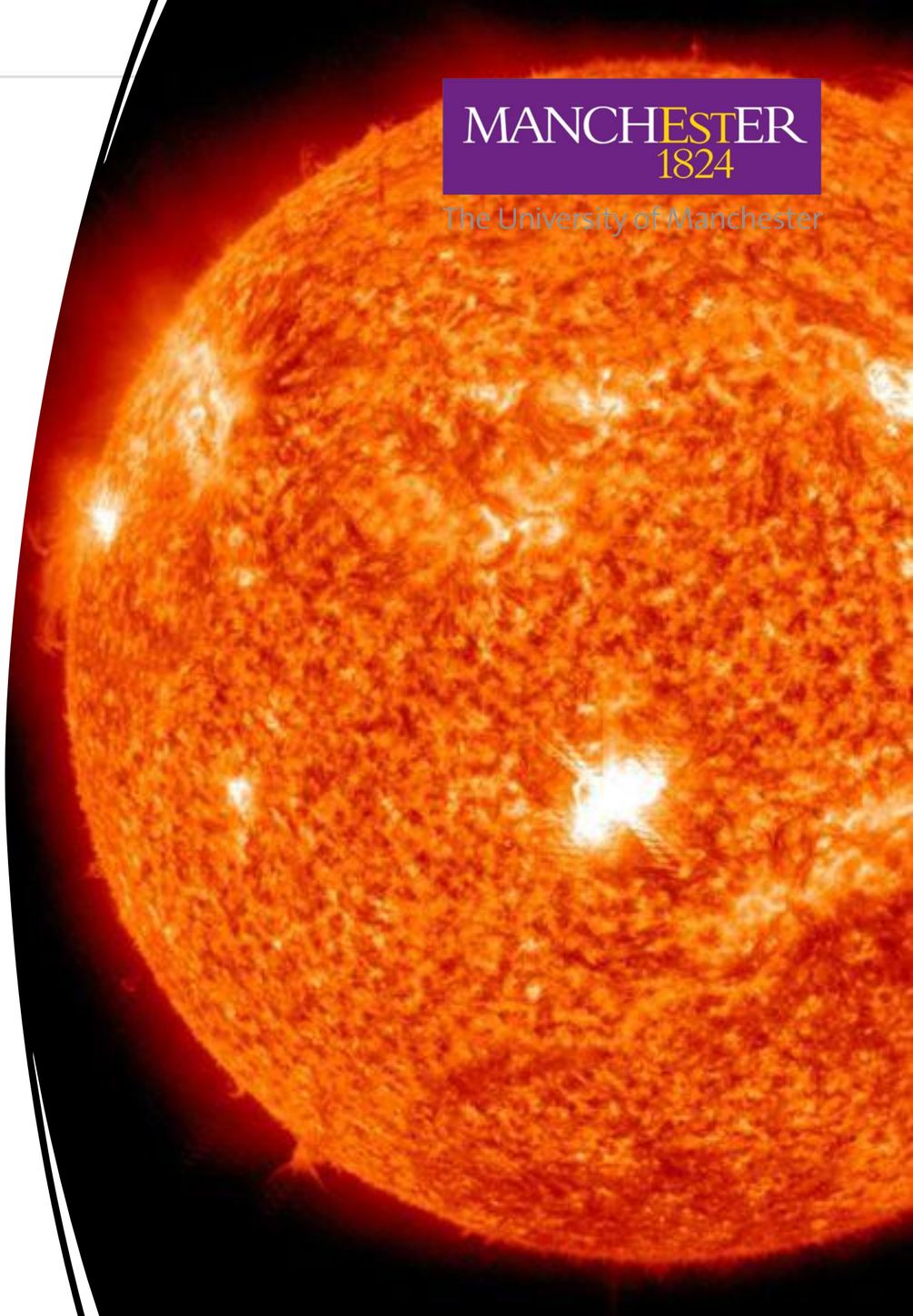
Solar output changes lead to:

- Changes to the atmospheric chemistry; decadal heating cycles; prolonged periods of cooler or warmer climates



Variations in Solar Activity







MANCHESTER 1824
The University of Manchester

Our key question today: *what drives natural variability in climate?*

1. Orbital Variations

2. Solar Variability

3. Volcanic Activity

4. Ocean Circulation

Sun – Earth System and Orbital changes

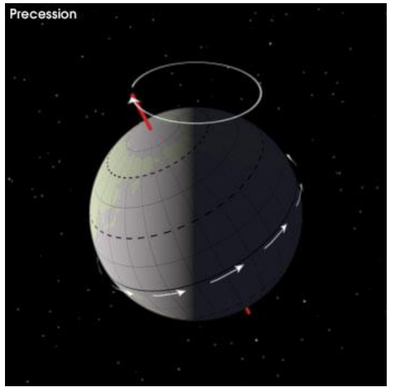
Milankovitch theory: Serbian astrophysicist in 1920's who studied effects of solar radiation on the irregularity of ice ages

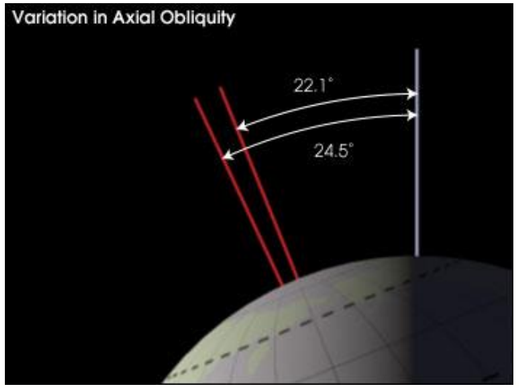
Variations in the Earth's orbit:

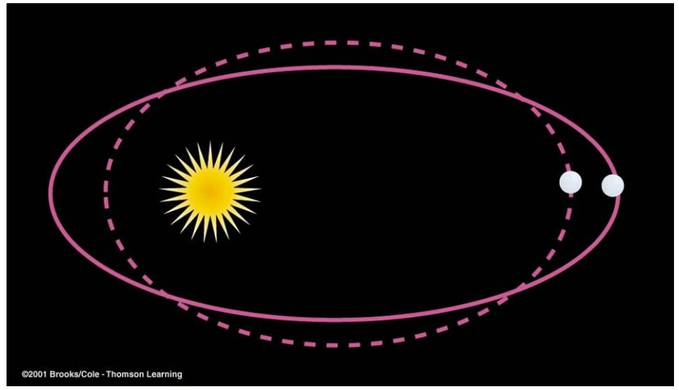
- Changes in shape of the earth's orbit around sun:
  - ***Eccentricity*** ***(100,000 years)***
- Wobbling of the earth's axis of rotation:
  - ***Precession*** ***(22,000 years)***
- Changes in the tilt of earth's axis:
  - ***Obliquity*** ***(41,000 years)***

When some parts of the cyclic variations are combined and occur at the same time, they are responsible for major changes to the earth's climate (e.g. ice ages).

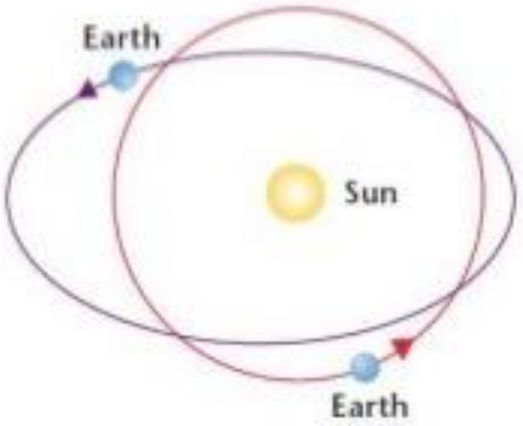


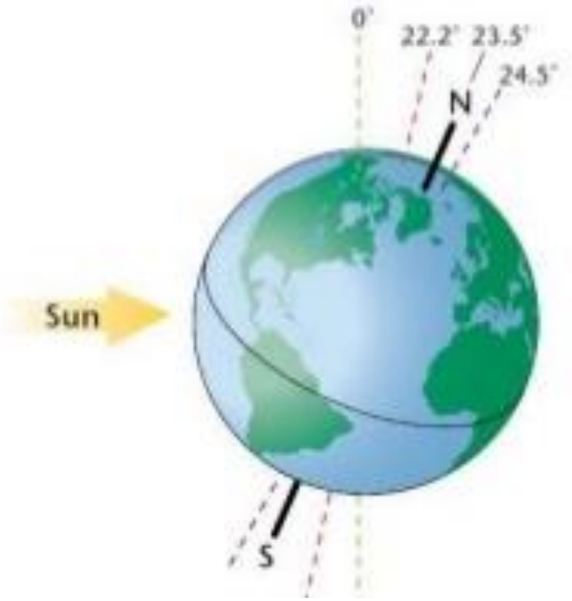






The Milankovitch Cycles







1. Eccentricity

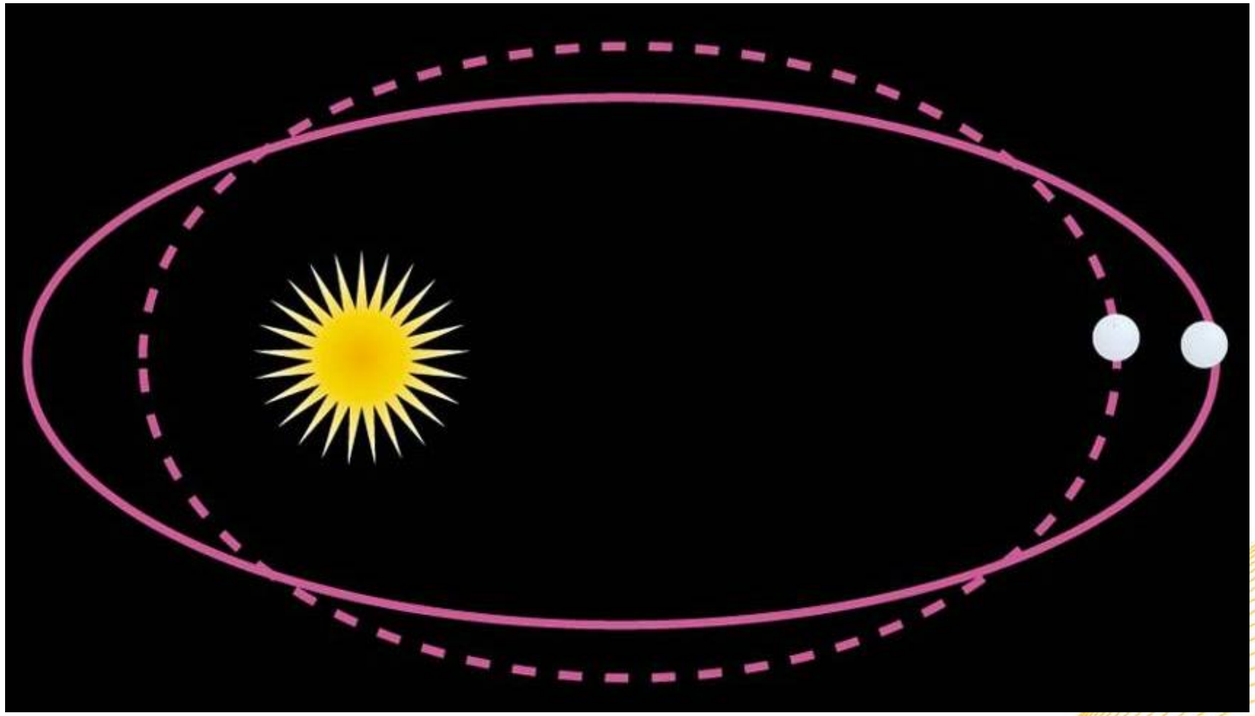
2. Obliquity

3. Precession

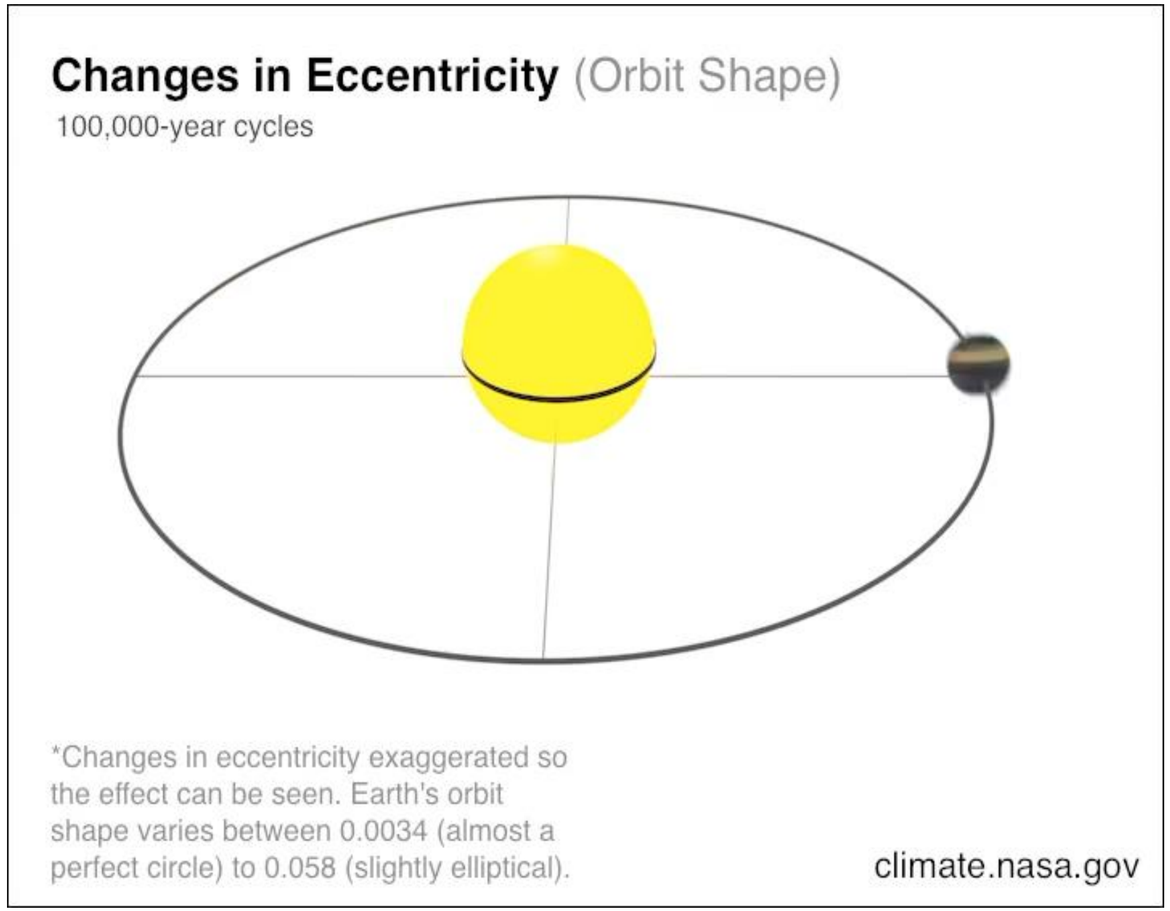
- Milankovitch cycles interact to modulate the amount, distribution, and seasonality of solar radiation reaching Earth's surface, contributing to long-term climate variability
- These cyclical orbital movements, cause variations of up to 25 percent in the amount of incoming insolation at Earth’s mid-latitudes

Eccentricity

- Eccentricity refers to the shape of Earth's orbit around the Sun, specifically how elongated or circular it is.
- Earth's orbit is not perfectly circular but slightly elliptical.
- Eccentricity varies cyclically over time, with periods typically ranging from **tens of thousands to hundreds of thousands of years.**
- They are primarily driven by the gravitational interactions between Earth and the gas giant planets, particularly Jupiter and Saturn.

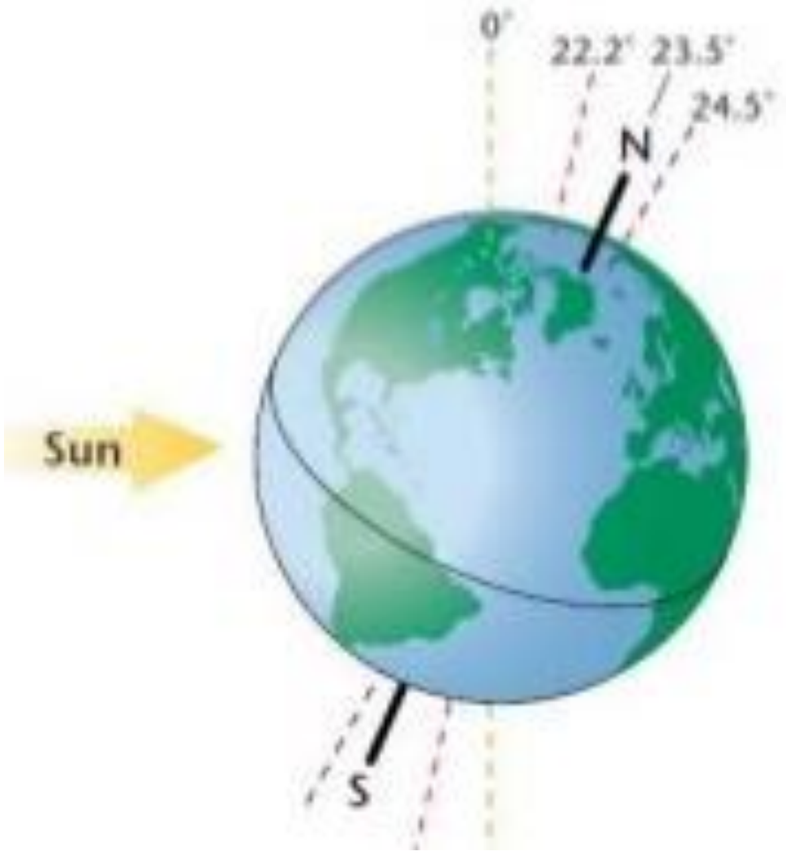


Eccentricity



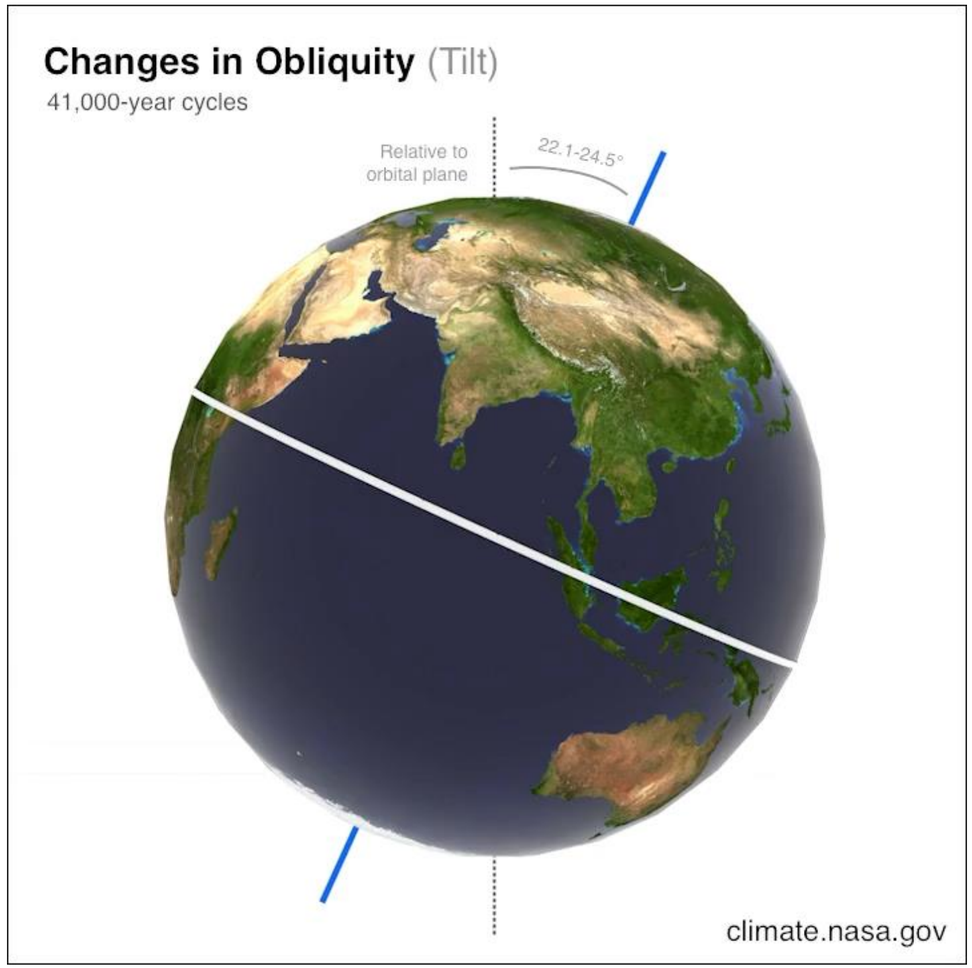
- When Earth's orbit is at its most elliptic, about 23 percent more incoming solar radiation reaches Earth at our planet's closest approach to the Sun each year than does at its farthest departure from the Sun.
- Currently, Earth's eccentricity is very slowly decreasing and is approaching its least elliptic (most circular), in a cycle that spans about 100,000 years.
- The minimum and maximum eccentricities during each 100,000 years also vary.
- Because variations in Earth's eccentricity are fairly small, they're a relatively minor factor in annual seasonal climate variations.

Obliquity



- Obliquity refers to the tilt of Earth's axis relative to its orbital plane around the Sun.
- Earth's axis is currently tilted at an angle of approximately 23.5 degrees relative to its orbital plane.
- This is not constant; it varies between approximately 22.1 and 24.5 degrees
- Over a cycle lasting about 41,000 years.
- Changes in obliquity affect the distribution of solar radiation across Earth's surface, leading to changes in the intensity and seasonality of climate patterns.

Obliquity

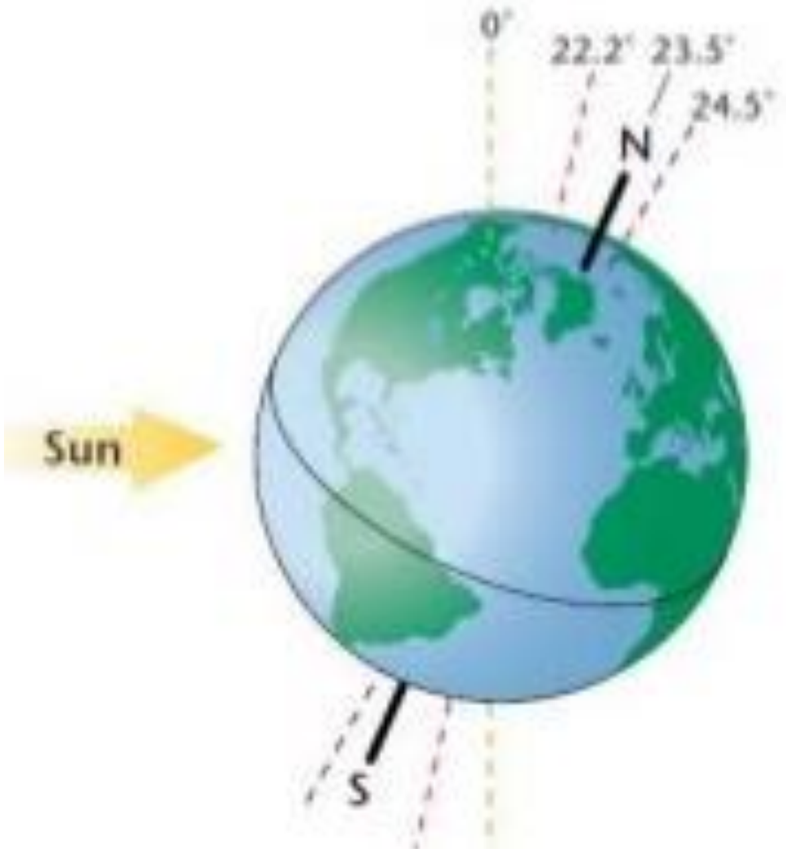


The obliquity of the Earth's axis of rotation is the angle of the axis relative to the line perpendicular to the plane of the Earth's orbit around the sun

Greater tilt = greater difference between seasons

Lesser Tilt = less of a difference between seasons

Obliquity



At the moment the obliquity is 23.5°, in the middle of the cycle.

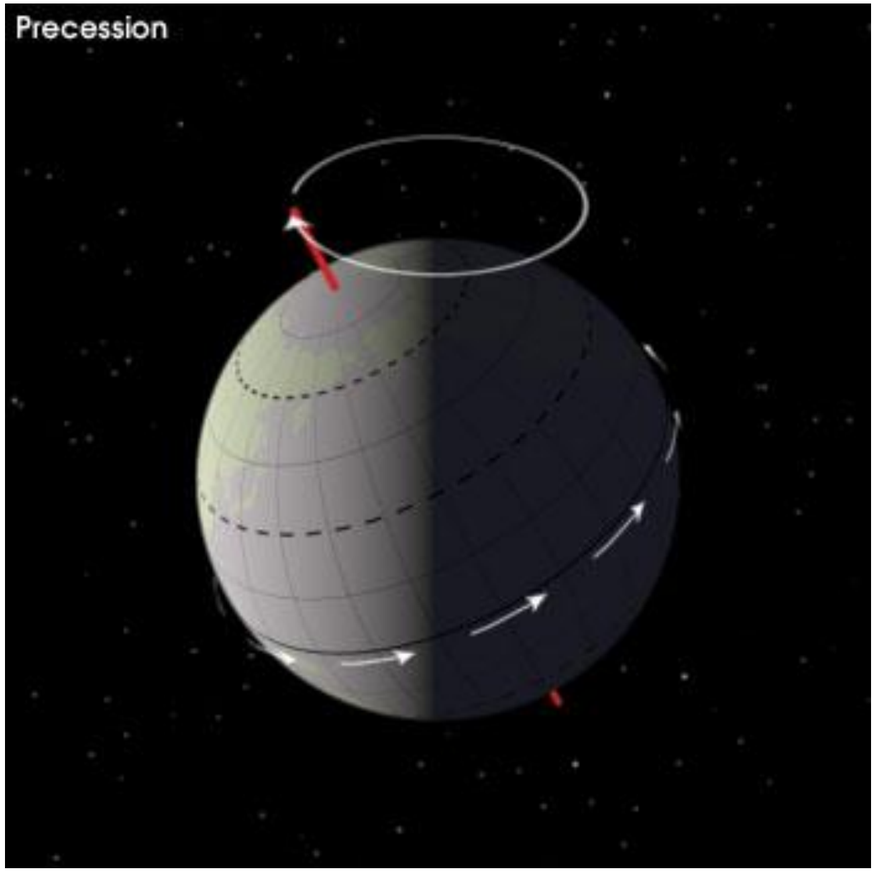
When the obliquity is low (22°) more sunlight intercepts the equator, increases the N-S temperature contrast.

When the obliquity is high (24.5°) more sunlight reaches higher latitudes, reducing temperature contrasts.

The obliquity therefore affects the seasons and temperature at each latitude.

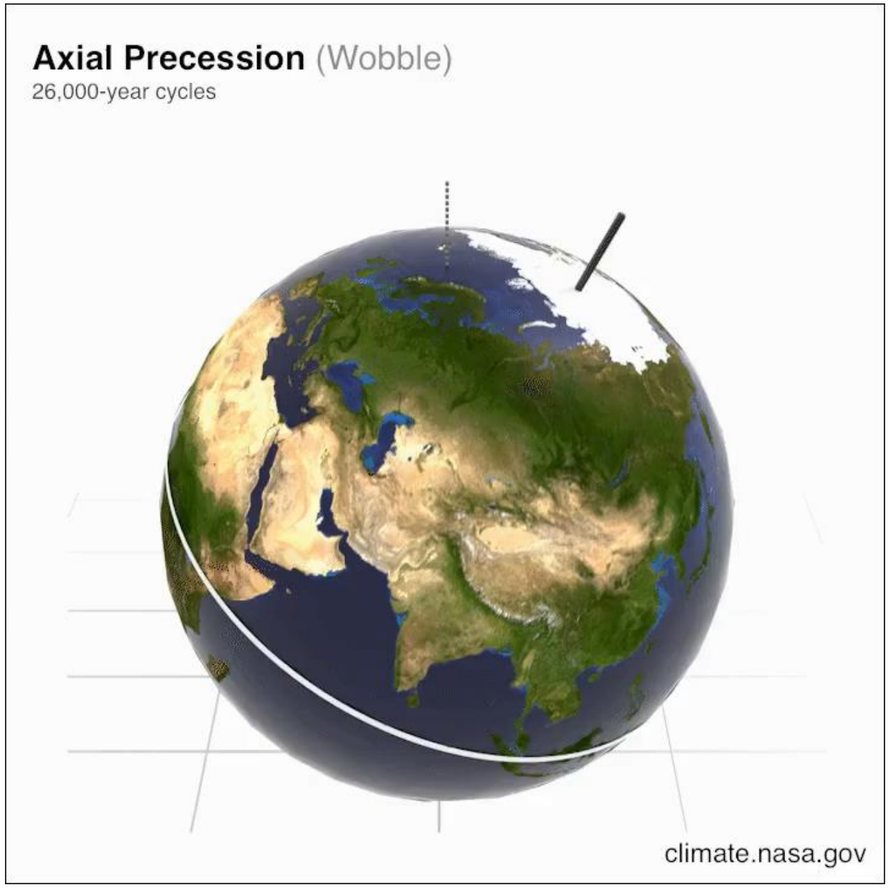
The change in Earth's obliquity, or axial tilt, is primarily driven by gravitational interactions between Earth and the Moon

Precession



- Precession refers to the gradual wobble or "precession" of Earth's rotational axis over time
- Like the way a spinning top wobbles as it slows down.
- This wobbling motion causes changes in the orientation of Earth's axis relative to its orbital path around the Sun.
- Precession operates over a period of 26,000 years
- These precessional cycles influence the timing of Earth's seasons and the distribution of solar radiation across different latitudes, impacting climate patterns.

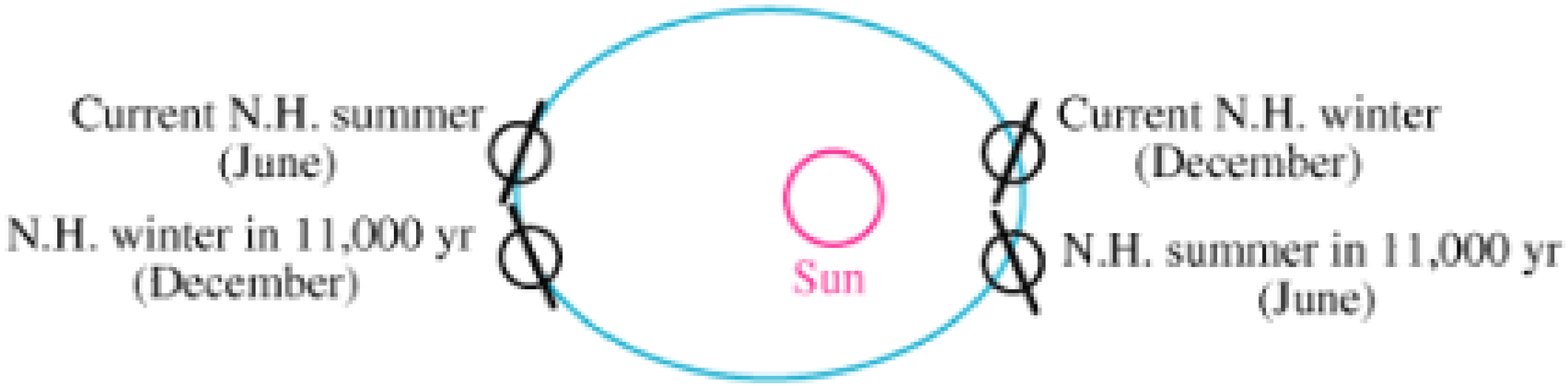
Precession



- Precession refers to the gradual wobble or "precession" of Earth's rotational axis over time
- Like the way a spinning top wobbles as it slows down.
- This wobbling motion causes changes in the orientation of Earth's axis relative to its orbital path around the Sun.
- Precession operates over a period of 26,000 years
- These precessional cycles influence the timing of Earth's seasons and the distribution of solar radiation across different latitudes, impacting climate patterns.

Precession

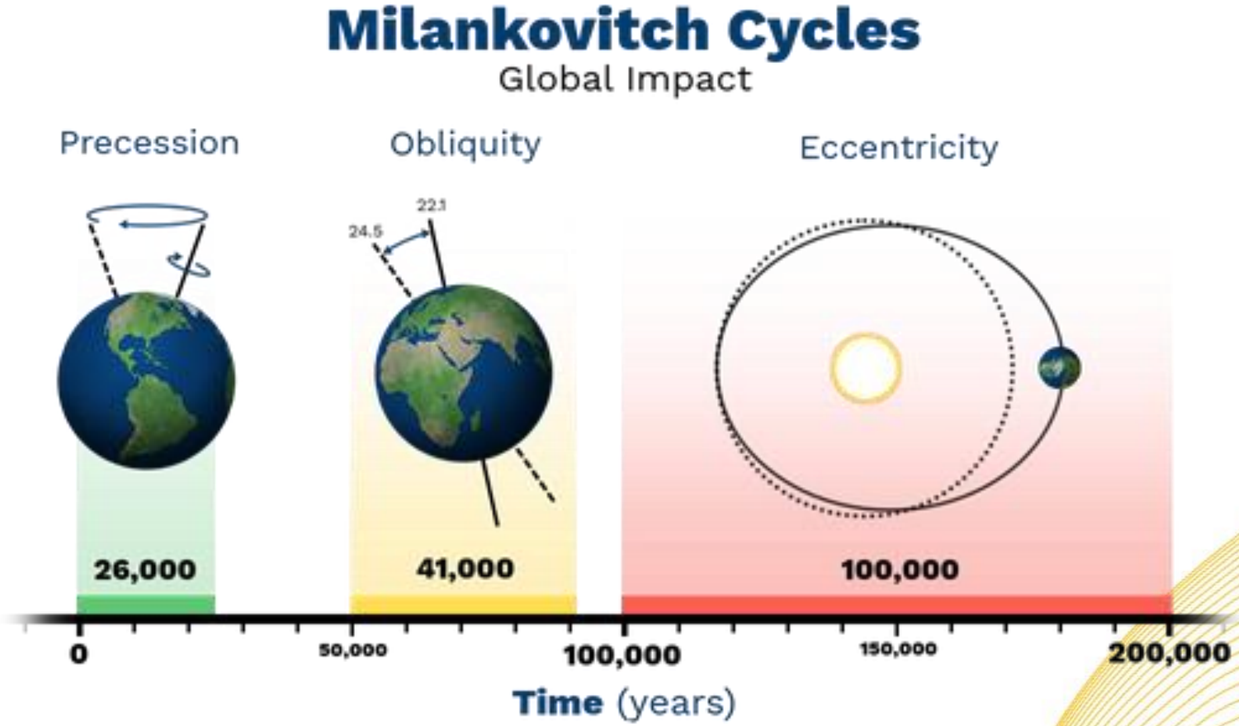
This is caused by the gravitational attraction between the Earth and other bodies in the solar system.



Precession does NOT change the yearly GLOBALLY AVERAGED incident solar radiation received by the Earth.
Instead it changes the quantity of incident radiation at each latitude during a season.
In 11,000 years e.g. NH summers will be warmer and SH summers will be cooler than they are today.

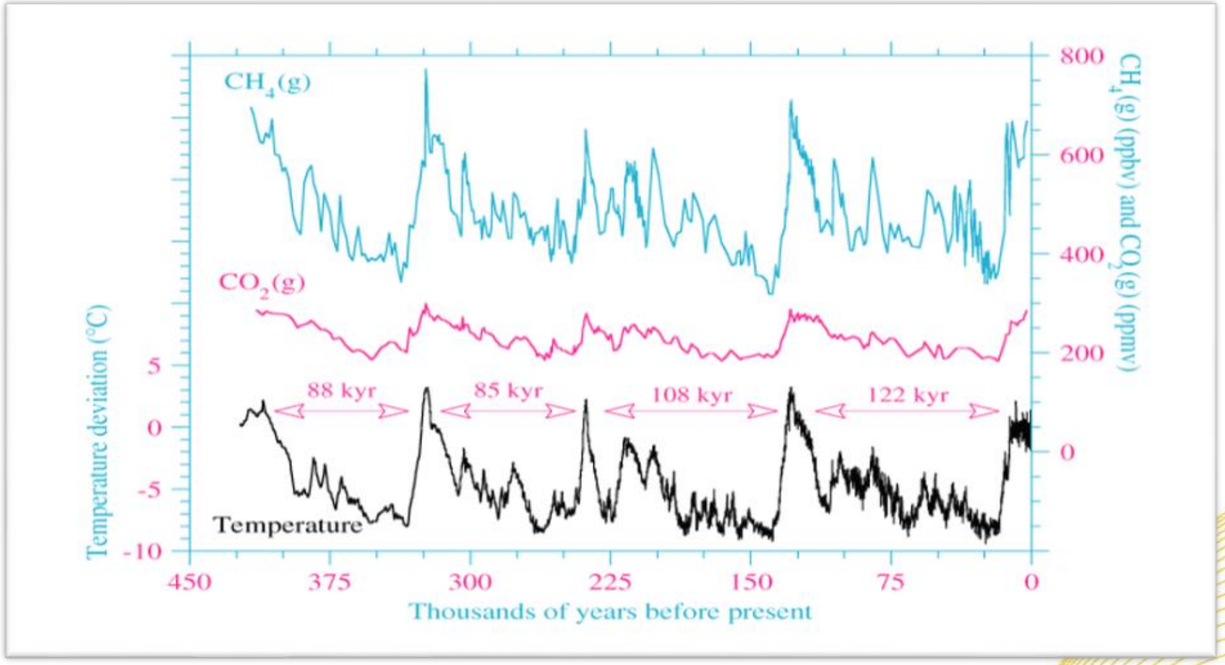
Milankovitch Cycles – The Theory

- Milankovitch developed a mathematical model combining orbital cycles to calculate changes in radiation by latitude and resulting surface temperatures
- The model works like a *climate time machine*, allowing climate conditions to be explored backward and forward in time
- Early calculations suggested Ice Ages occurred roughly every **41,000 years**
- Ice Age pacing later shifted (~800,000 years ago) to a **100,000-year cycle**, matching Earth's eccentricity
- The reason for this shift is still not fully understood



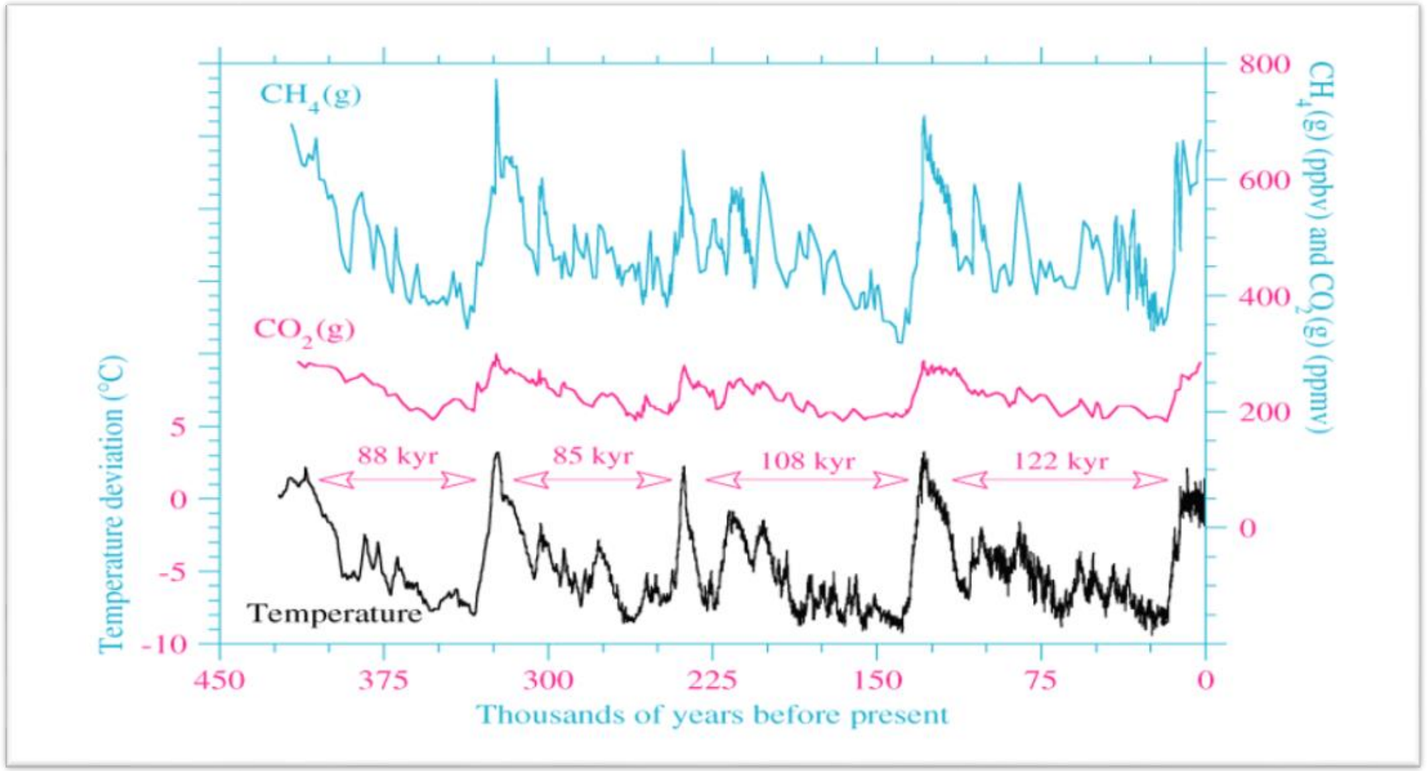
Milankovitch Cycles – Verification

- Initially overlooked, Milankovitch's theory gained recognition after his death
- **1976 Science study (Hays et al.)** linked orbital cycles to climate change using deep-sea sediment cores
- Temperature records reconstructed for the past **450,000 years**
- Ice Ages coincided with specific stages of Earth's orbital variation
- Ice-core data from Greenland and Antarctica further confirmed the cycles
- Milankovitch cycles are now widely accepted as drivers of glacial–interglacial timing



Discussion (5 mins)

- In past climate records, does temperature change before or after CO_2?
- What starts the change first: orbital changes, temperature, or CO_2?
- From prior knowledge what drives these observed changes in CO_2?





MANCHESTER 1824
The University of Manchester

Our key question today: *what drives natural variability in climate?*

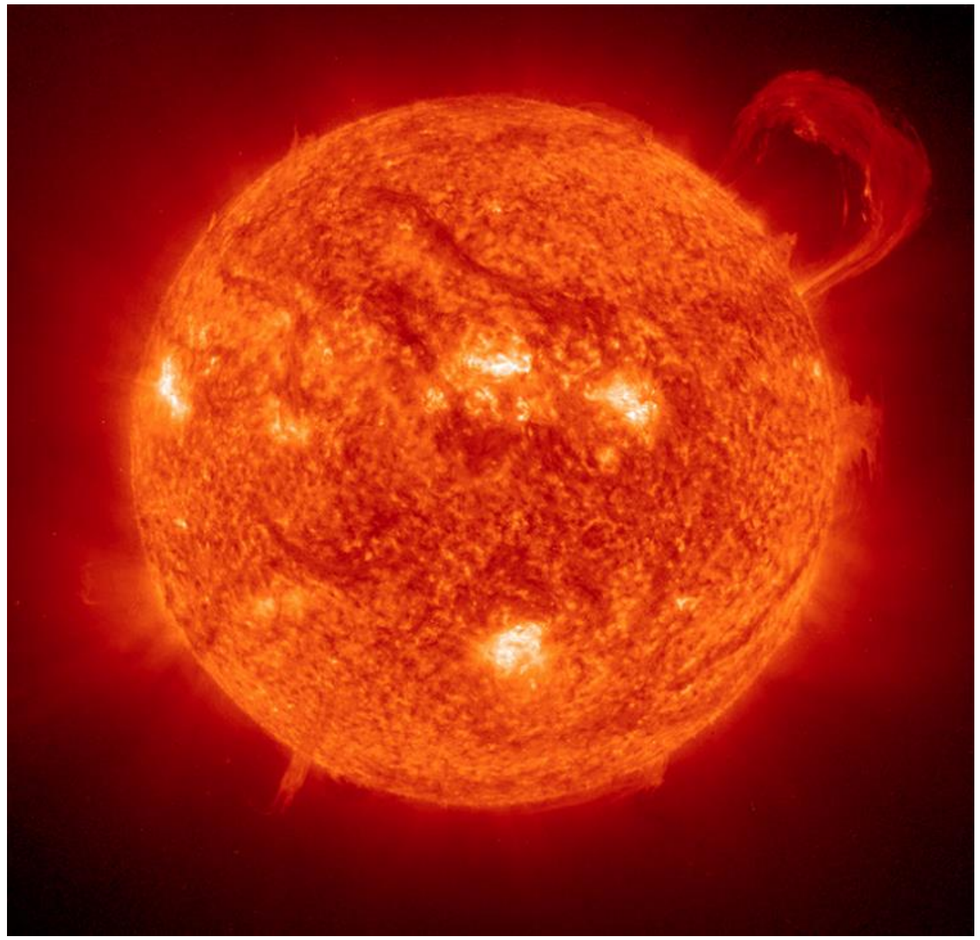
1. **Orbital Variations**

2. **Solar Variability**

3. **Volcanic Activity**

4. **Ocean Circulation**

Solar Output Changes



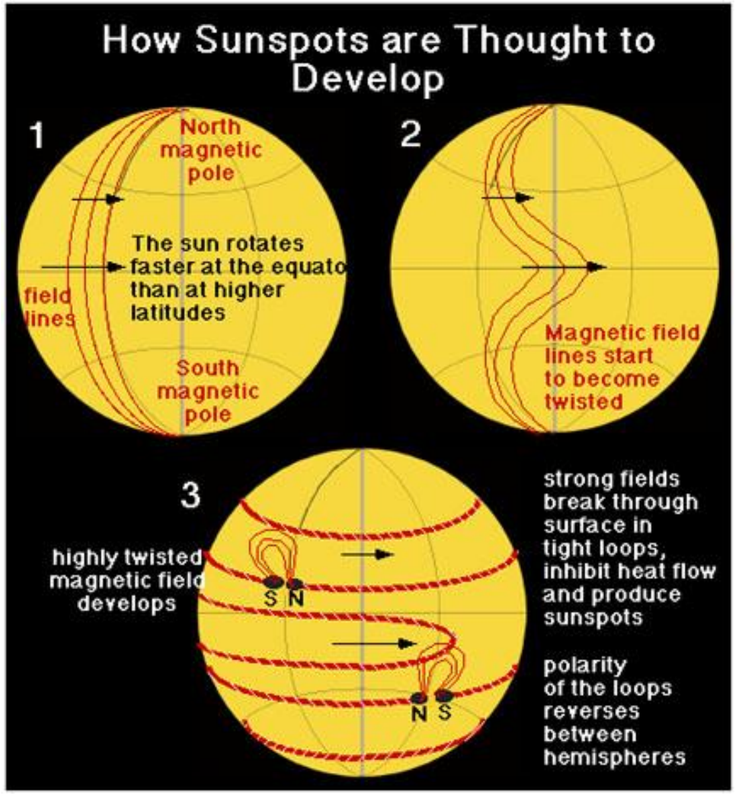
Through its lifetime, the Sun naturally goes through changes in energy output.

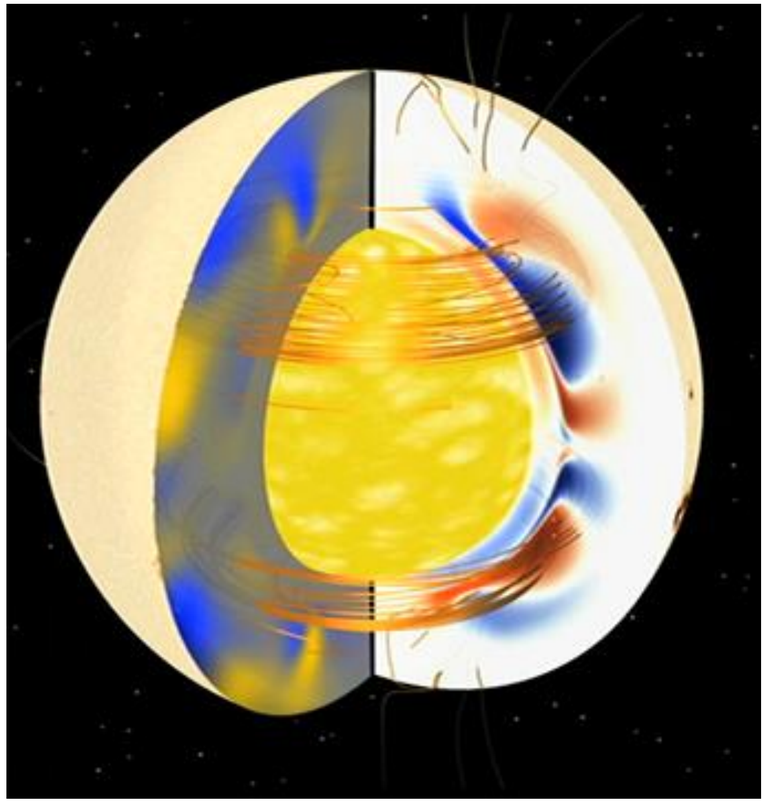
Some of these occur over a regular 11-year period of peak (many sunspots) and low activity (fewer sunspots), which are quite predictable.

During strong cycles, the Sun's total brightness at solar maximum is about 0.1 percent higher than it is at solar minimum.

Sunspots are areas that appear as temporary spots that are darker than the surrounding areas

Mechanisms in the Sun





Sun's poles swap polarity every 11-years – a 22 year magnetic cycle

During this switch there is prevention of convection of hot core to surface

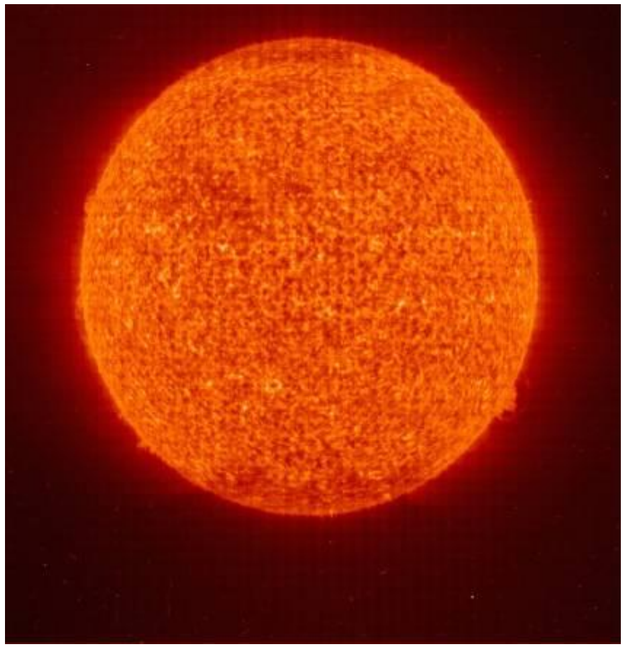
Spots appear dark because they are cooler than surface

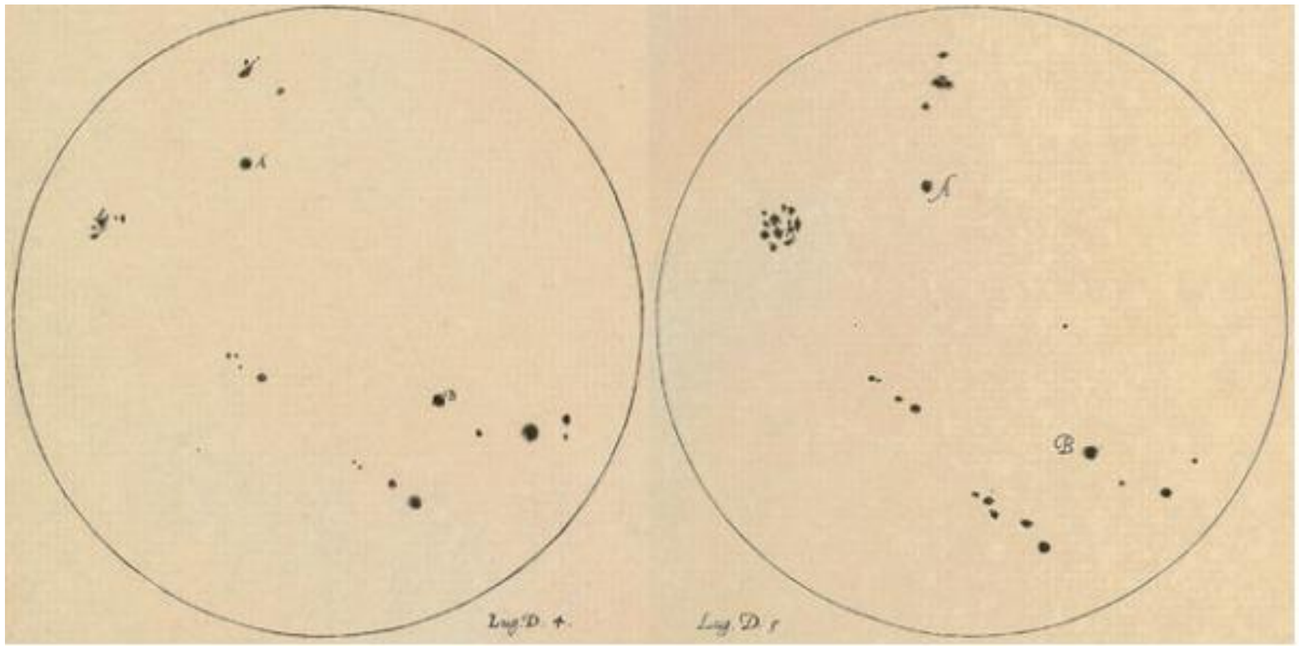
Bright areas around spots are emitting more at shorter-wavelengths

Solar Variations – Sunspots

Sunspots correlate with solar activity

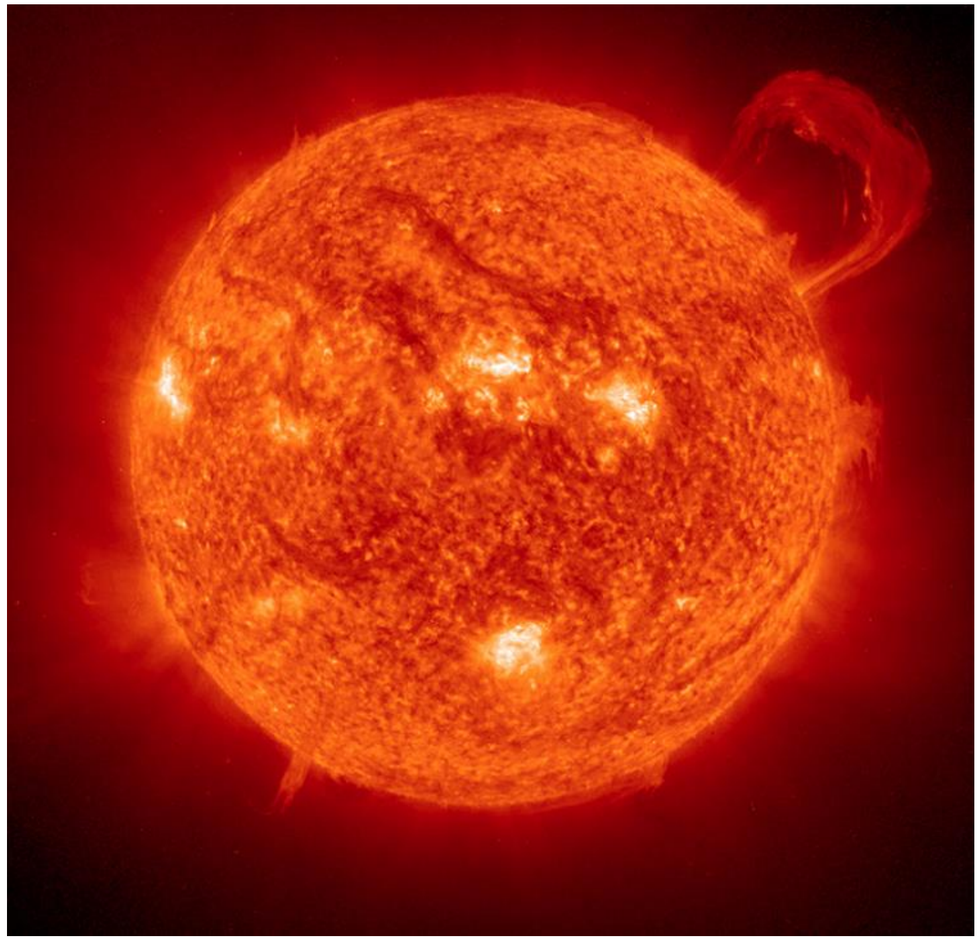
more sunspots, more solar energy





Drawings by Galileo, showing sunspots he observed through a telescope on July 1613.

Possible Climatic Effects



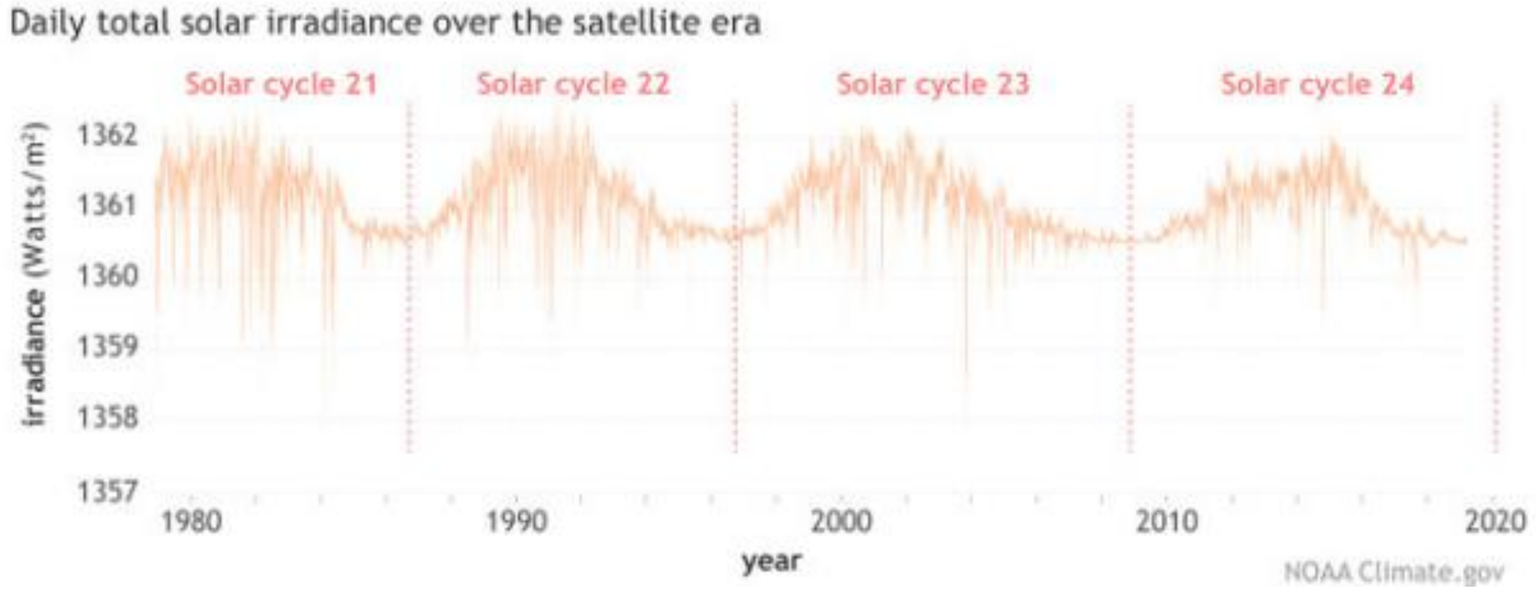
More shortwave radiation incident on the earth …..warming

Increases in absorption of radiation at particular wavelengths…..warming

Increases in ozone produced by extra UV radiation…warms stratosphere

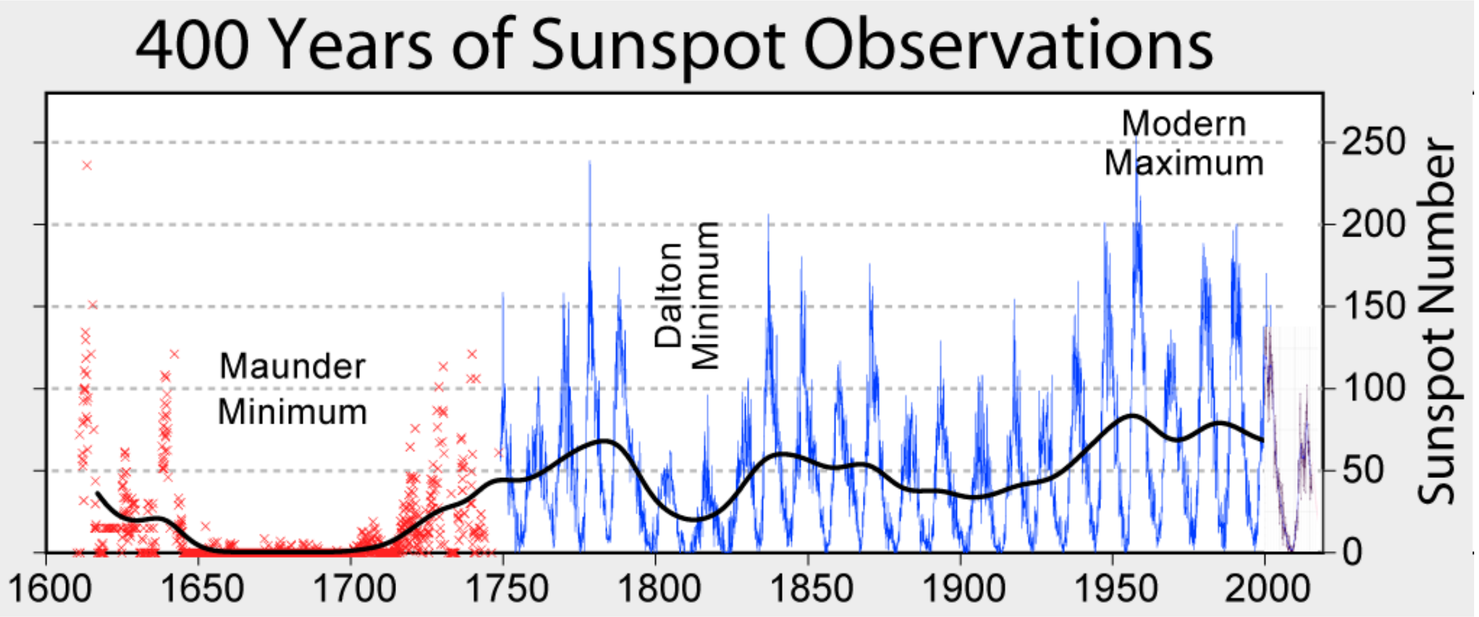
The 11 Year Solar Cycle

- The sun exhibits significant changes over the 11-year solar cycle.
- All types of solar energy output are modulated by this cycle.
- The sun can appear dramatically different over much shorter timescales than 11 years; some wavelengths can vary drastically in minutes.



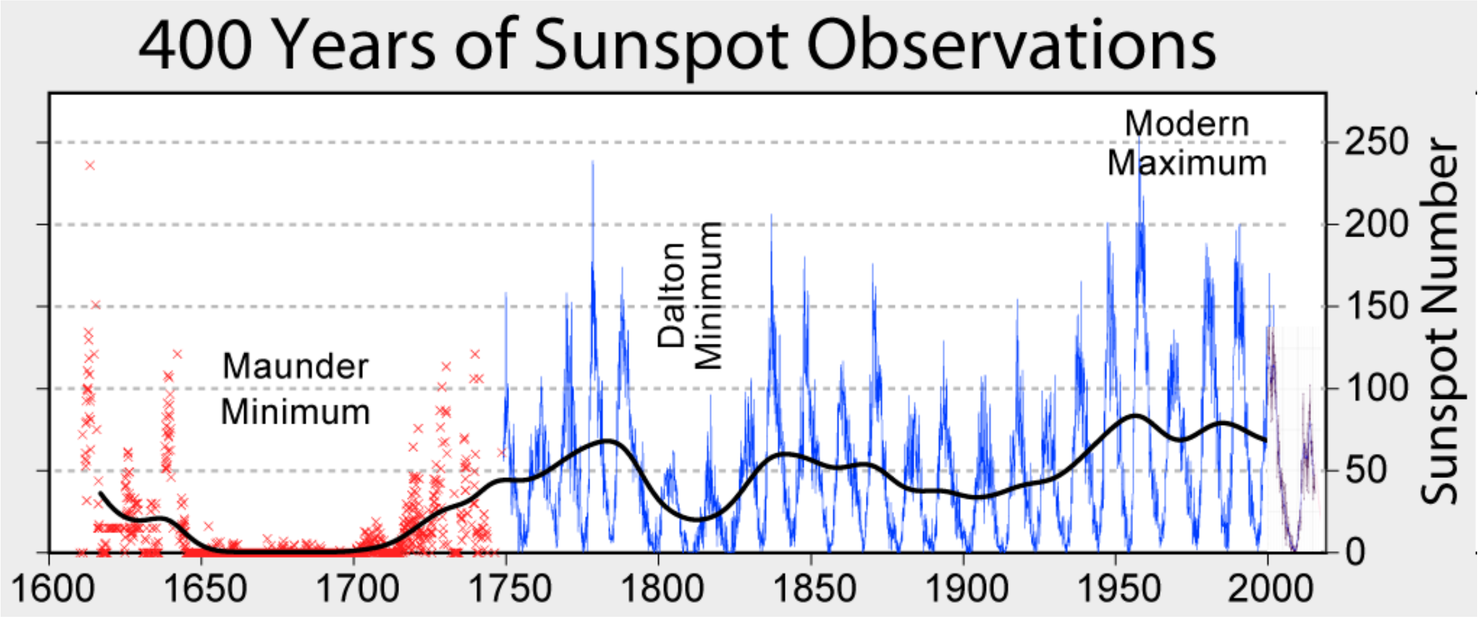
Daily observations of total solar irradiance (orange line) since the start of the satellite era in 1978. Day-to-day, TSI may vary by as much as 0.3 percent, but average differences between maximum and minimum are on the order 0.1 percent, or around 1 Watt per square meter.

The 11 Year Solar Cycle



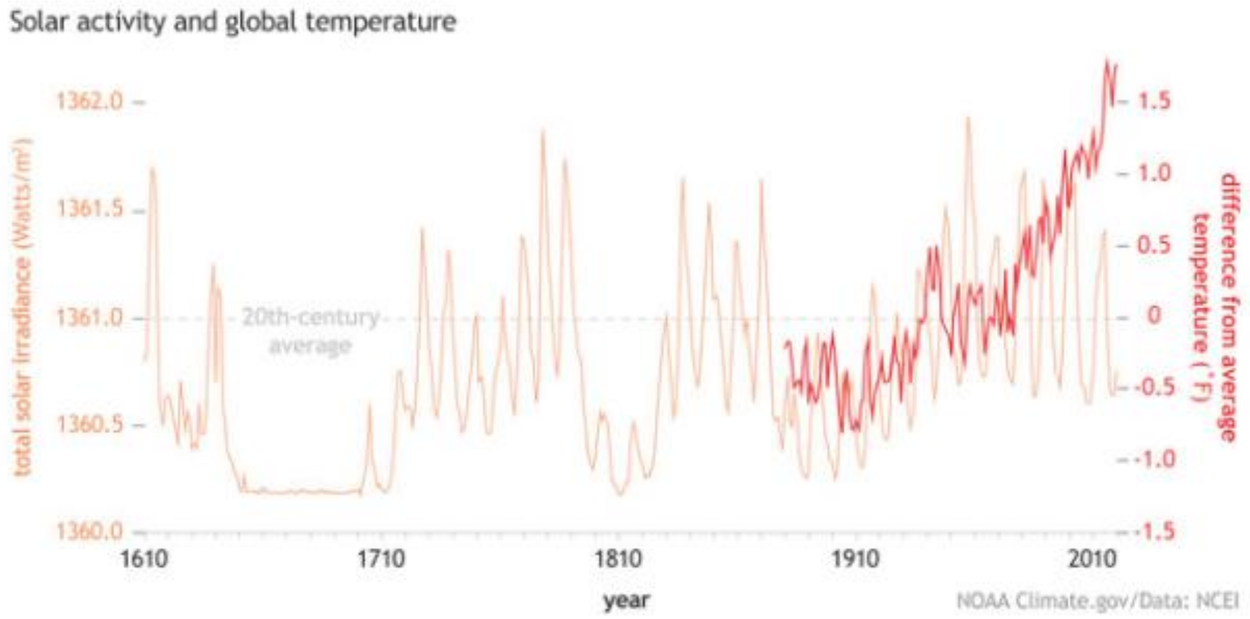
- Grand Solar Minimums: Periods when sunspots virtually disappear for decades
- Example: Maunder Minimum (1645-1715)
- Characteristics: Sunspots nearly absent
- Overlapped with Little Ice Age (1450-1850)
- Little Ice Age: Centuries-long cold spell, strongest in Northern Hemisphere

The Maunder Minimum



- An absence of sunspots was well observed from 1645 to 1715.
- The so-called “Maunder minimum” coincided with a cool climatic period in Europe and North America: “Little Ice Age”

How do changes in solar output effect climate?



Yearly total solar irradiance (orange line) from 1610–2020 and the annual global temperature compared to the 20th-century average (red line) from 1880–2020. Since the middle of the 20th-century, solar activity has declined while global temperature increased rapidly.

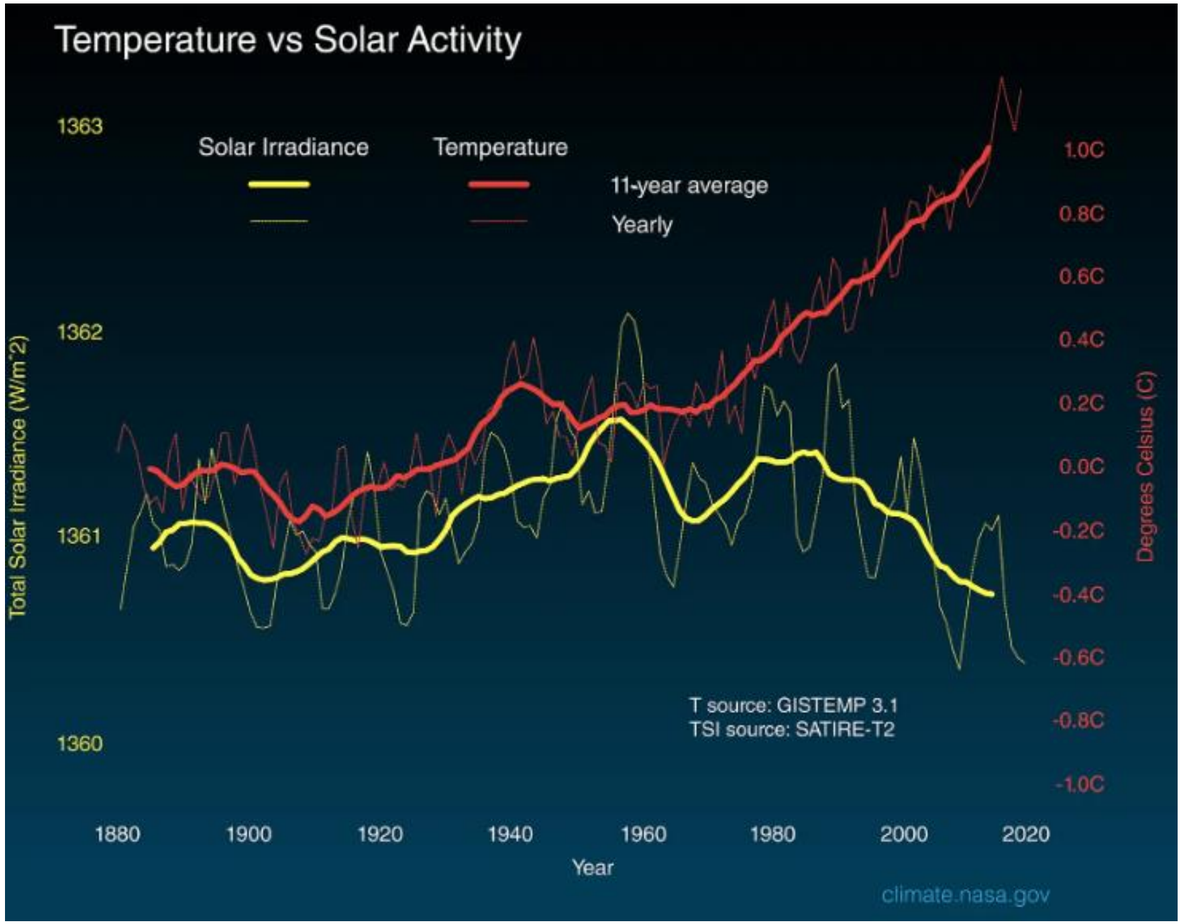
IPCC research findings:

- Average solar cycle brightness change: 0.06 Watts per square meter
- Estimated contribution to warming: Around 0.01 degrees Celsius
- Percentage of total warming attributed: Approximately 1%

Global warming over industrial era:

- Range: 0.95–1.2 degrees Celsius (2011–2020 versus 1850–1900)
- Solar influence on this warming: Minimal, around 1%

Is the sun causing global warming?



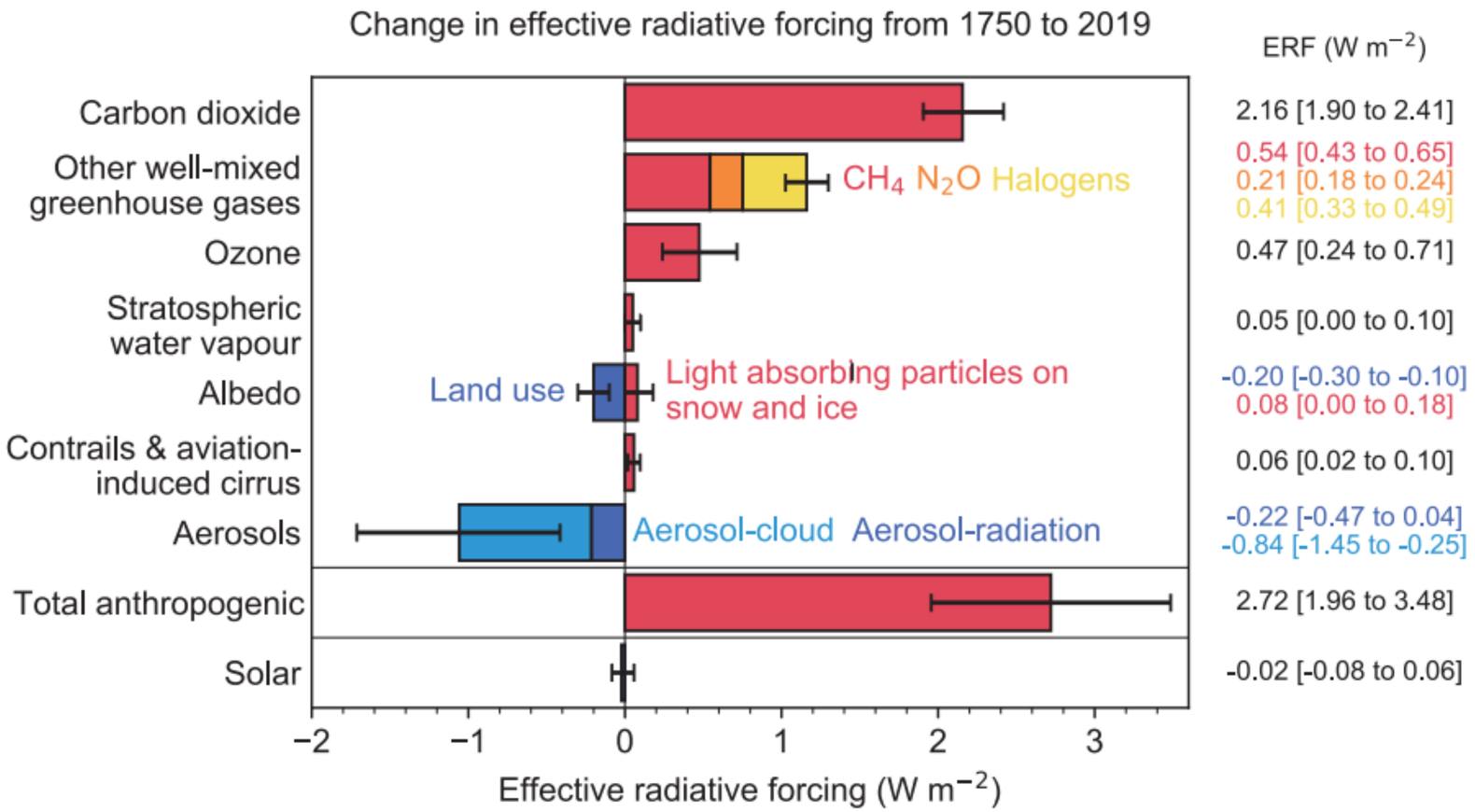
Since 1978, we have been tracking the amount of solar energy that hits the top of the atmosphere using sensors on satellites, which tell us that there has been no upward trend in the amount of solar energy reaching our planet.

If the Sun were responsible for global warming, we would expect to see warming throughout all layers of the atmosphere, from the surface to the upper atmosphere (stratosphere).

But what we actually see is warming at the surface and cooling in the stratosphere. This is consistent with the warming being caused by a buildup of heat-trapping gases near Earth's surface, and not by the Sun getting “hotter.”

Since 1750, the warming driven by greenhouse gases coming from the human burning of fossil fuels is over 270 times greater than the slight extra warming coming from the Sun itself over that same time interval.

Is the sun causing global warming?





MANCHESTER 1824
The University of Manchester

Our key question today: *what drives natural variability in climate?*

1. **Orbital Variations**

2. **Solar Variability**

3. **Volcanic Activity**

4. **Ocean Circulation**

Volcanic Climate Influence



Cooling agents:

Sulphur dioxide turning into sulphuric acid droplets (reflective aerosol)

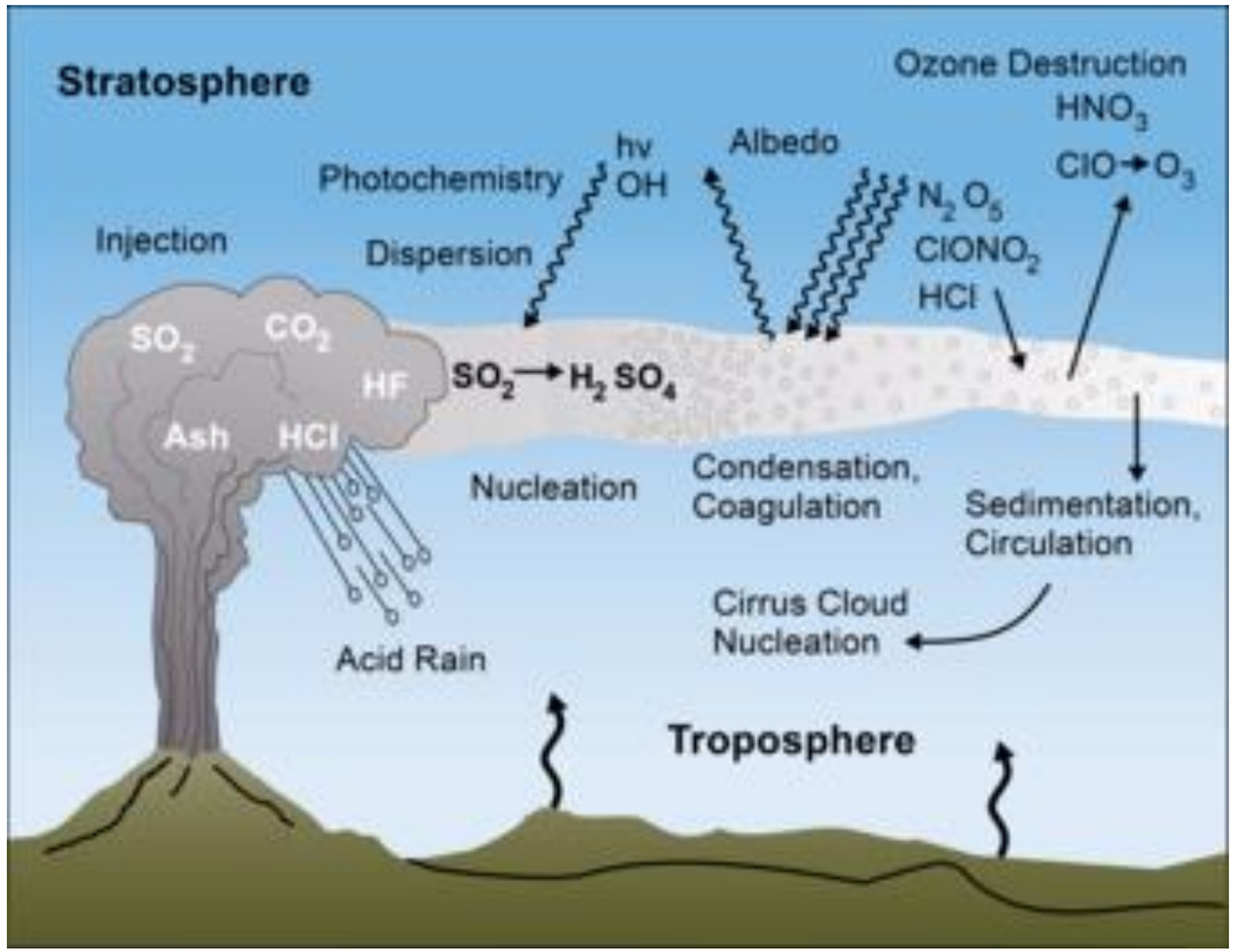
Microparticles of dust that absorb and scatter solar radiation in the stratosphere

Impacts:

Global average temperatures affected the following year following a major eruption

Hemispheric impacts can be even greater

Volcanic Climate Influence

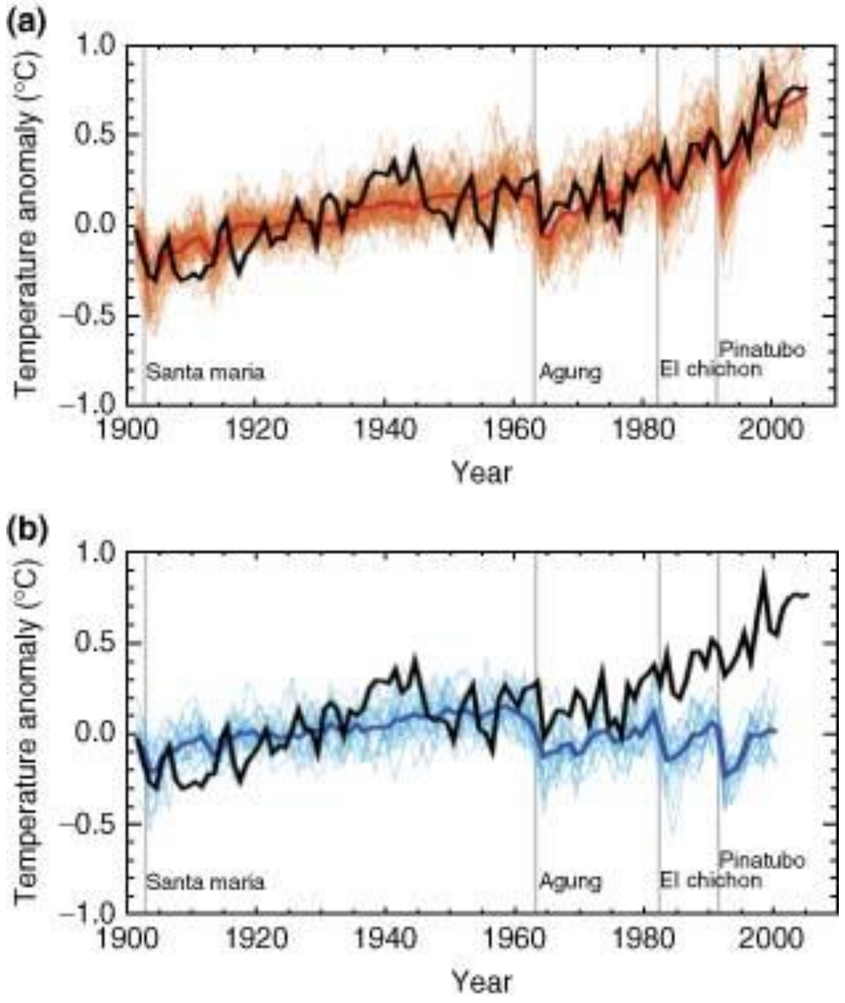


The most significant climate impacts from volcanic injections into the stratosphere come from the conversion of sulfur dioxide to sulfuric acid, which condenses rapidly in the stratosphere to form fine sulfate aerosols.

The aerosols increase the reflection of radiation from the Sun back into space, cooling the Earth's lower atmosphere or troposphere.

CO_2 from volcanoes estimated to be only 1% of total emissions of CO_2

Volcanic Climate Influence

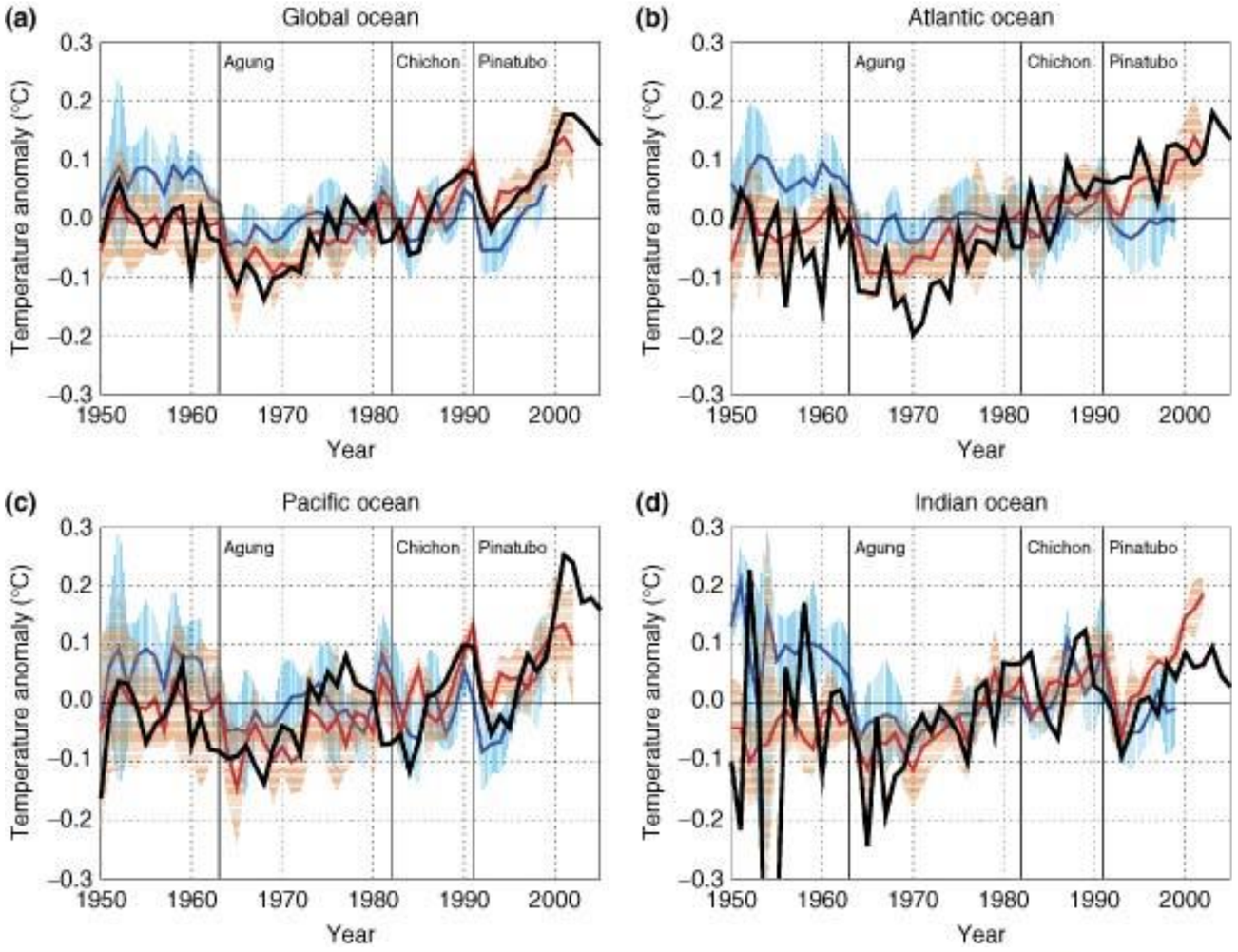


Global average temps reduced by several tenths of a degree following year

Would require succession of eruptions to have a lasting impact.

The eruption of Mount Pinatubo on June 15, 1991, was one of the largest eruptions of the twentieth century and injected a 20-million ton sulfur dioxide cloud into the stratosphere at an altitude of more than 20 miles.

Volcanic Climate Influence





MANCHESTER 1824
The University of Manchester

Our key question today: *what drives natural variability in climate?*

1. **Orbital Variations**

2. **Solar Variability**

3. **Volcanic Activity**

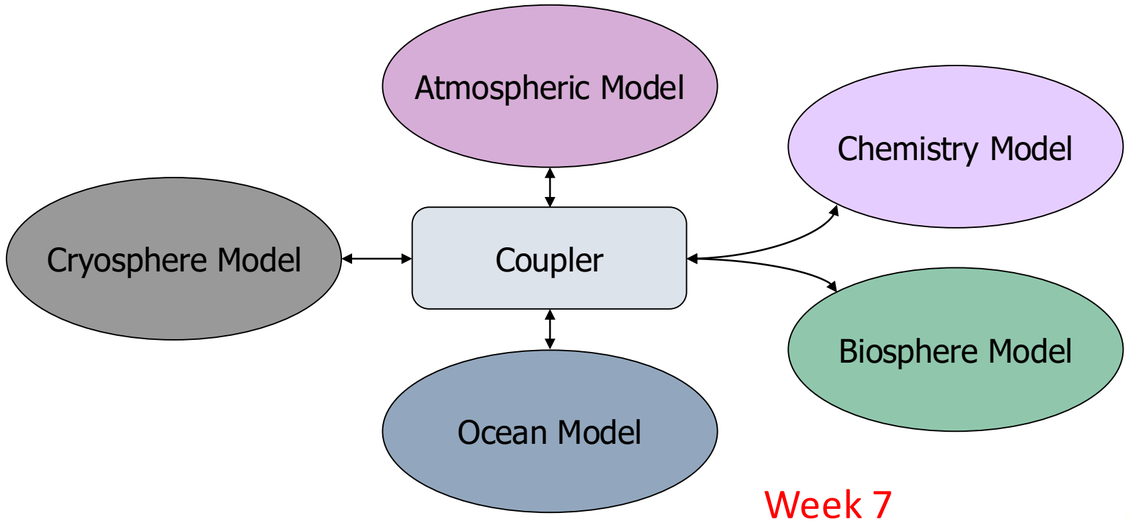
4. **Ocean Circulation**

The ENSO (El Nino/Southern Oscillation)

What is ENSO?

A coherent, large-scale fluctuation of ocean temperatures, rainfall, atmospheric circulation and air pressure across the tropical Pacific. It is a coupled ocean-atmosphere phenomenon.

ENSO is one of the most important climate phenomena on Earth due to its ability to change the global atmospheric circulation, which in turn, influences temperature and precipitation across the globe.



Week 7

The ENSO (El Nino/Southern Oscillation)

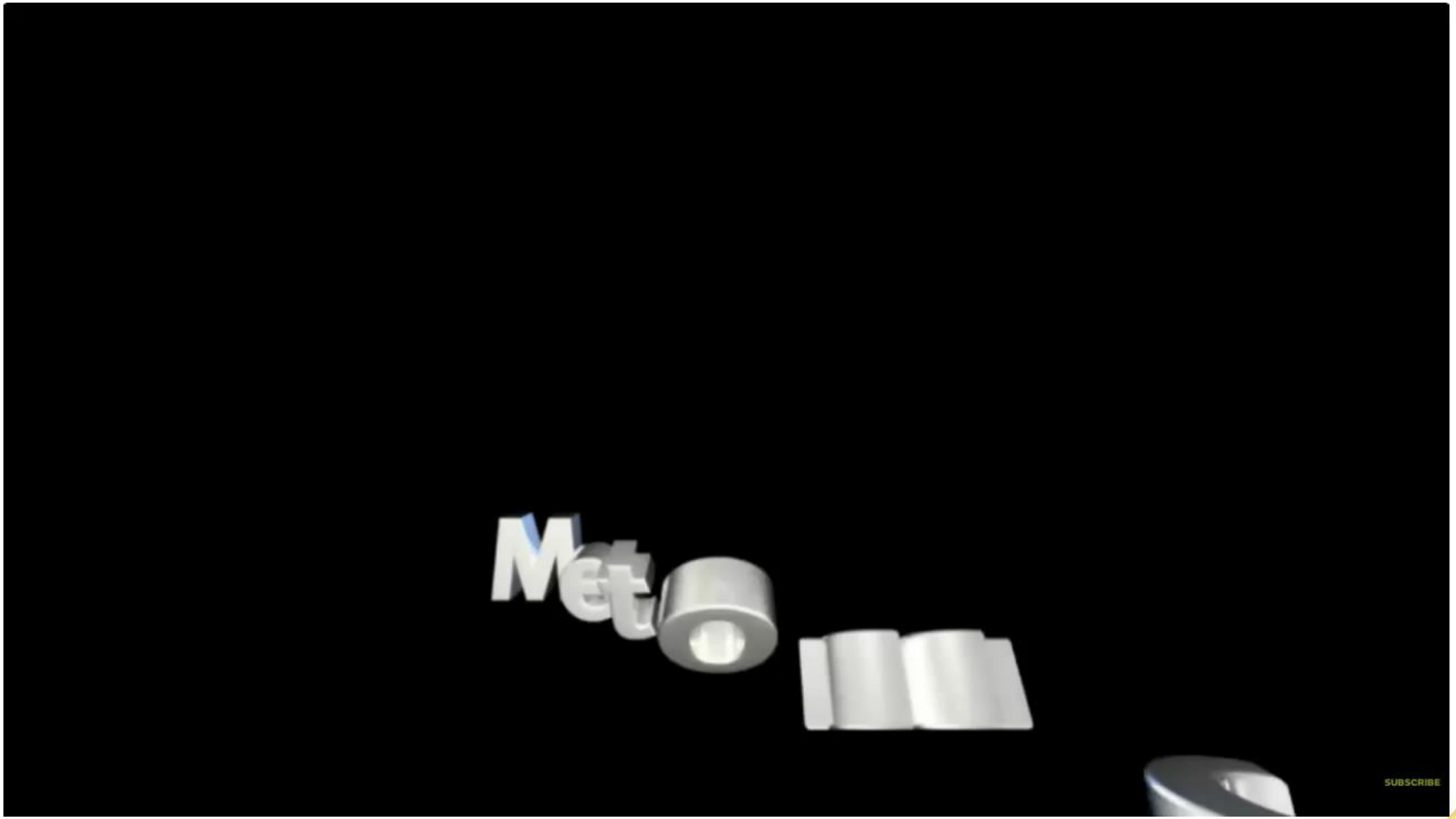
The El Niño-Southern Oscillation (ENSO) is a recurring climate pattern involving changes in the temperature of waters in the central and eastern tropical Pacific Ocean.

On periods ranging from about three to seven years, the surface waters across the tropical Pacific Ocean warm or cool by anywhere from 1°C to 3°C, compared to normal.

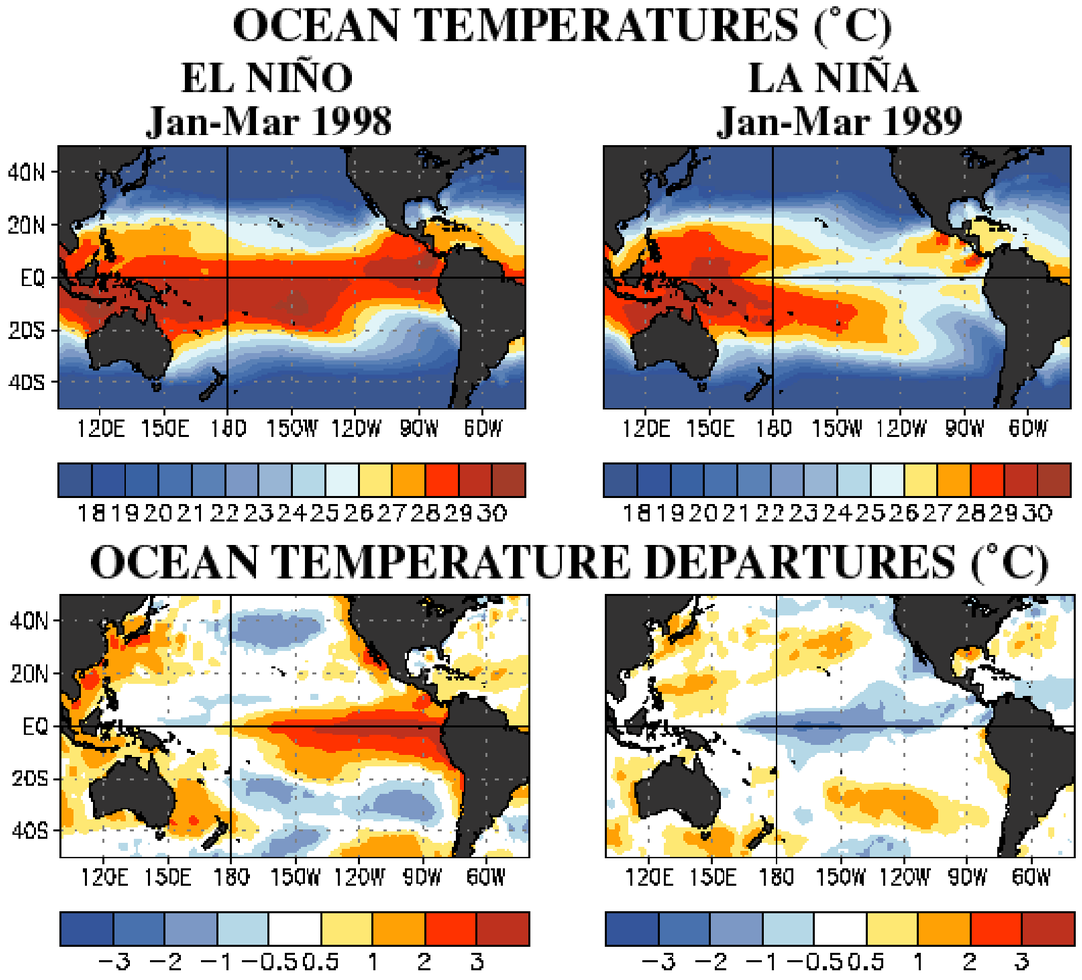
This oscillating warming and cooling pattern, referred to as the ENSO cycle, directly affects rainfall distribution in the tropics and can have a strong influence on weather across the United States and other parts of the world.

El Niño and La Niña are the extreme phases of the ENSO cycle; between these two phases is a third phase called ENSO-neutral.

The scale of these fluctuations is vast, changes in sea-surface temperatures often span a distance of more than one-quarter the circumference of the globe, and changes in tropical rainfall and winds span a distance of more than one-half the circumference of the earth.


SUBSCRIBE

The ENSO (El Nino/Southern Oscillation)



•***El Niño:*** A warming of the ocean surface in the central and eastern tropical Pacific Ocean. Over Indonesia, rainfall tends to become reduced while rainfall increases over the central and eastern tropical Pacific Ocean. During a strong El Niño episode, ocean temperatures can locally average 2°C - 5°C above normal

•***La Niña:*** A cooling of the ocean surface. Over Indonesia, rainfall tends to increase while rainfall decreases over the central and eastern tropical Pacific Ocean. During a strong La Niña episode ocean temperatures locally average 1°C – 4°below normal

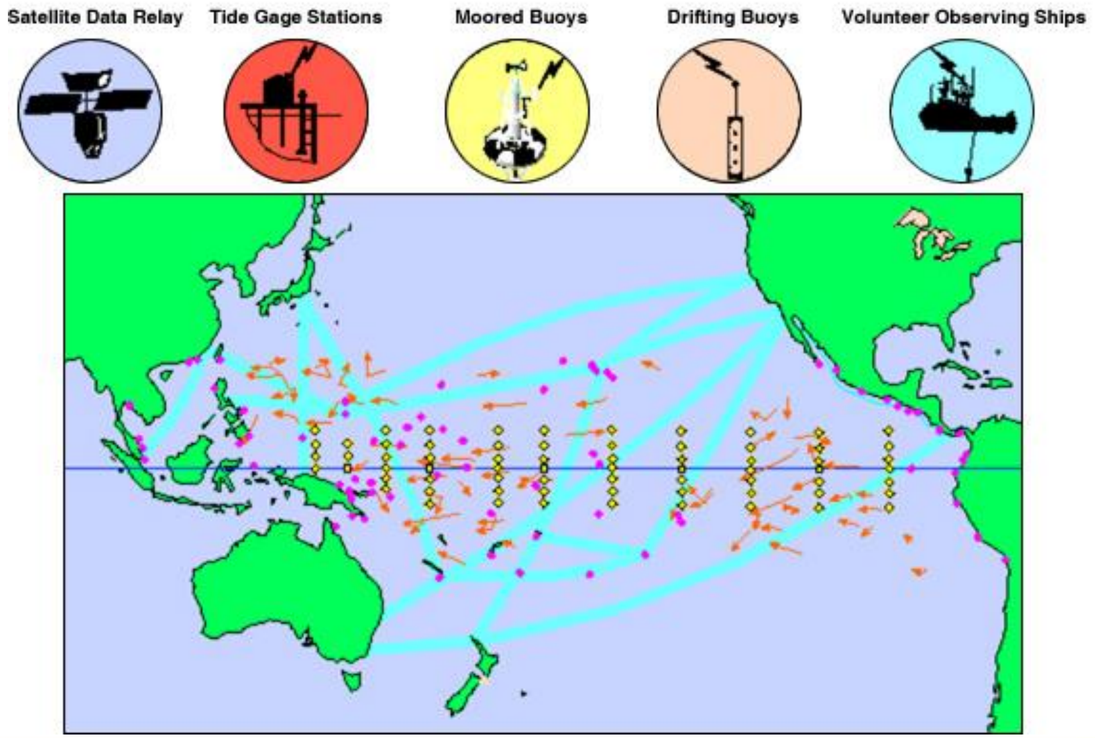
•***Neutral:*** Neither El Niño or La Niña. Often tropical Pacific ocean temps are generally close to average. However, there are some instances when the ocean can look like it is in an El Niño or La Niña state, but the atmosphere is not playing along (or vice versa).

During a strong El Niño episode, ocean temperatures can locally average 2°C - 5°C above normal between the date line and the west coast of South America, resulting in nearly uniform surface temperatures across the entire equatorial Pacific.

The Southern Oscillation Index (SOI)

The SOI is one measure of the large-scale fluctuations in air pressure occurring between the western and eastern tropical Pacific (i.e., the state of the southern Oscillation) during El Niño and La Niña episodes.

This index has been calculated based on the differences in air pressure anomaly between Tahiti and Darwin, Australia.



Data collected using a variety of observing platforms are used to construct various atmospheric and oceanic indices to monitor the ENSO cycle.

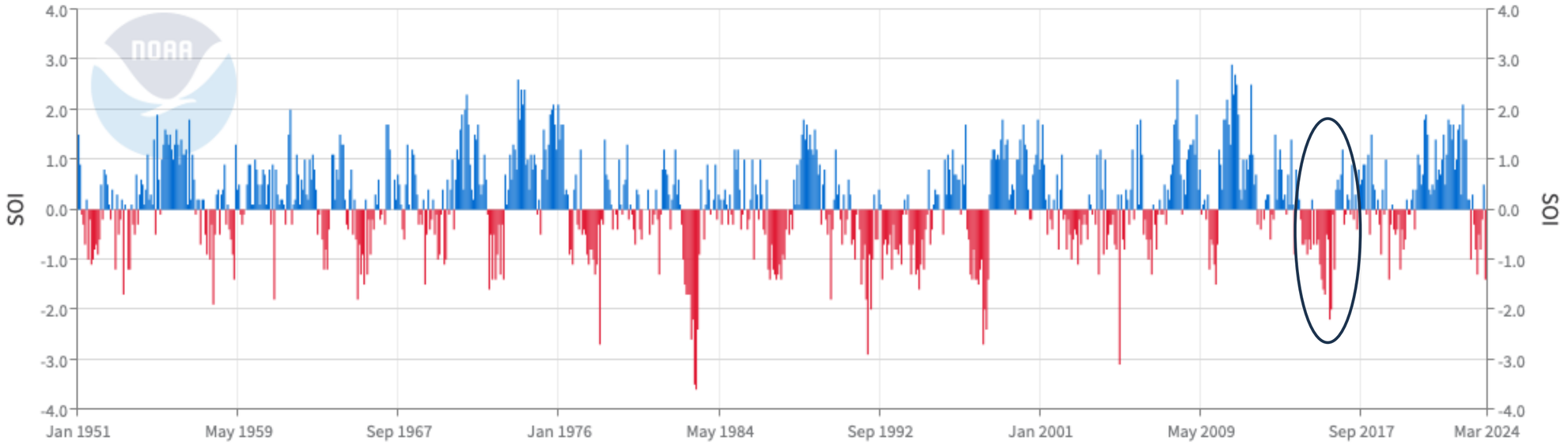
The commonly used atmospheric indices are:

- **The Tahiti Darwin SOI**
- **The equatorial SOI**
- **Low-level wind index**
- **Upper-level wind index**
- **Outgoing Longwave Radiation (OLR), a proxy for tropical precipitation**
- **A new Index, the Oceanic Niño Index (ONI)**

The Southern Oscillation Index (SOI)

The SOI is one measure of the large-scale fluctuations in air pressure occurring between the western and eastern tropical Pacific (i.e., the state of the southern Oscillation) during El Niño and La Niña episodes.

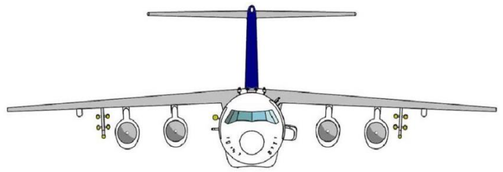
This index has been calculated based on the differences in air pressure anomaly between Tahiti and Darwin, Australia.



The Southern Oscillation Index (SOI)

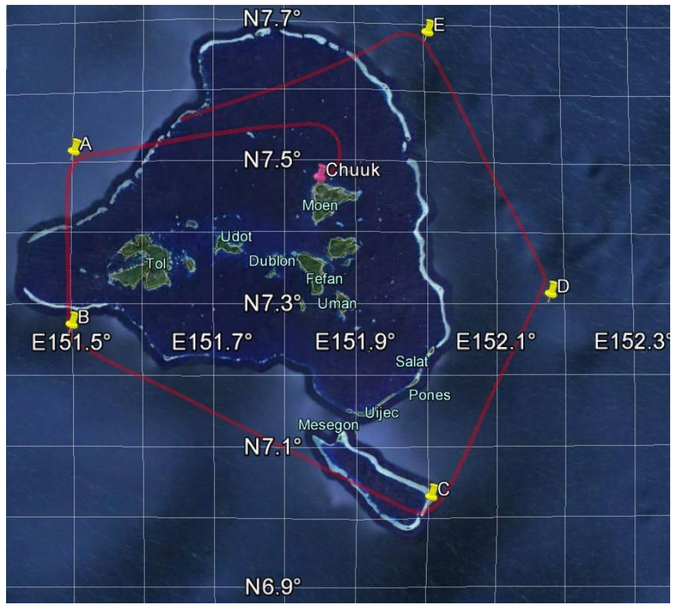
The SOI is one measure of the large-scale fluctuations in air pressure occurring between the western and eastern tropical Pacific (i.e., the state of the southern Oscillation) during El Niño and La Niña episodes.

This index has been calculated based on the differences in air pressure anomaly between Tahiti and Darwin, Australia.



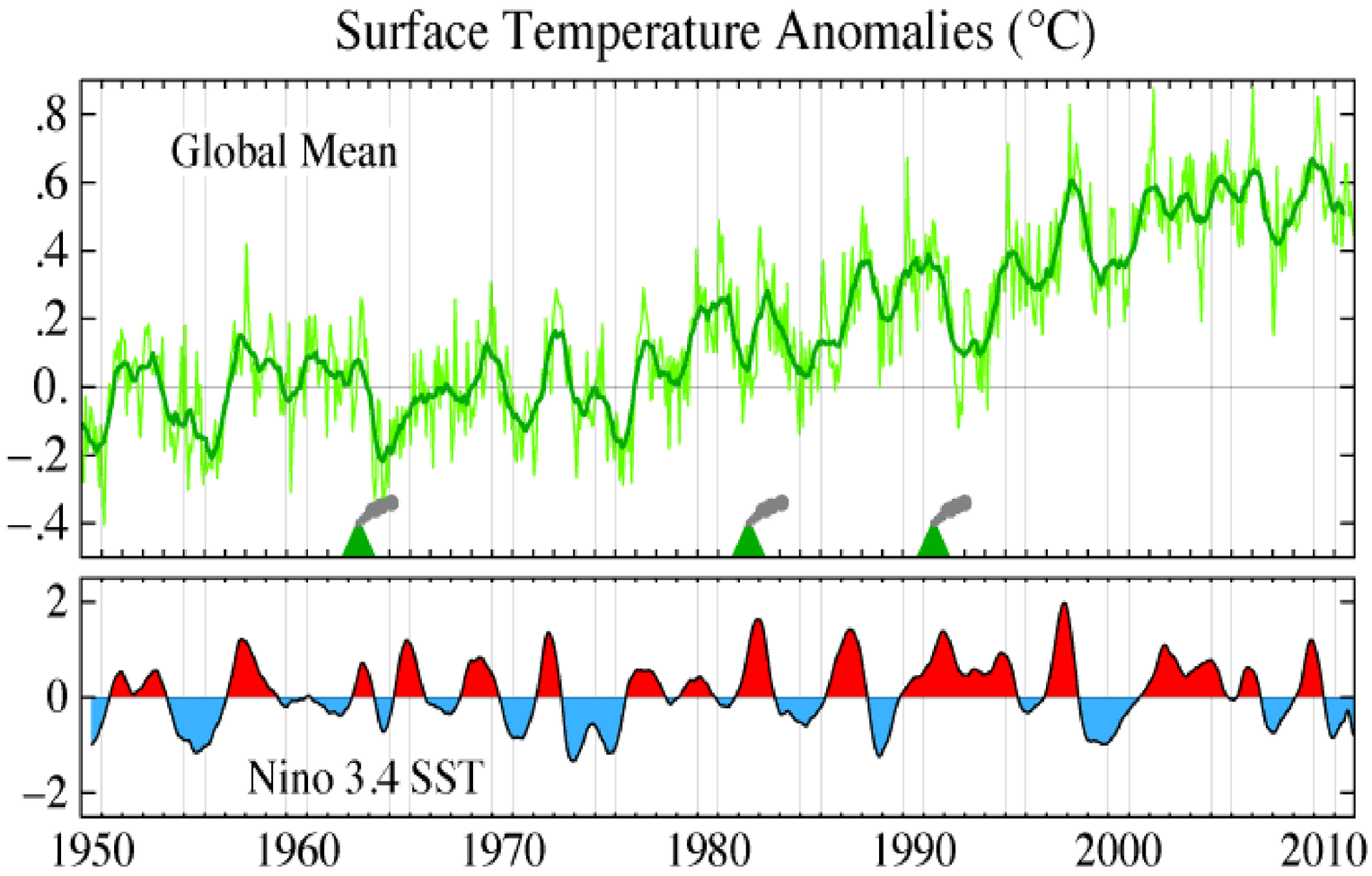
COORDINATED AIRBORNE STUDIES IN THE TROPICS







Impact of ENSO on Global Surface Temps



- El-Nino causes a warming of the surface temperature by a few tenths of a degree with a lag as heat need to be distributed around globe by weather systems
- El-Nino causes a warming of the middle and upper tropospheric temperatures by roughly the same amount but with a further delay as the heat has to be transferred upwards by weather systems
- La-Nina has similar cooling effects on the surface and with similar delays

Key Points of Natural Variability



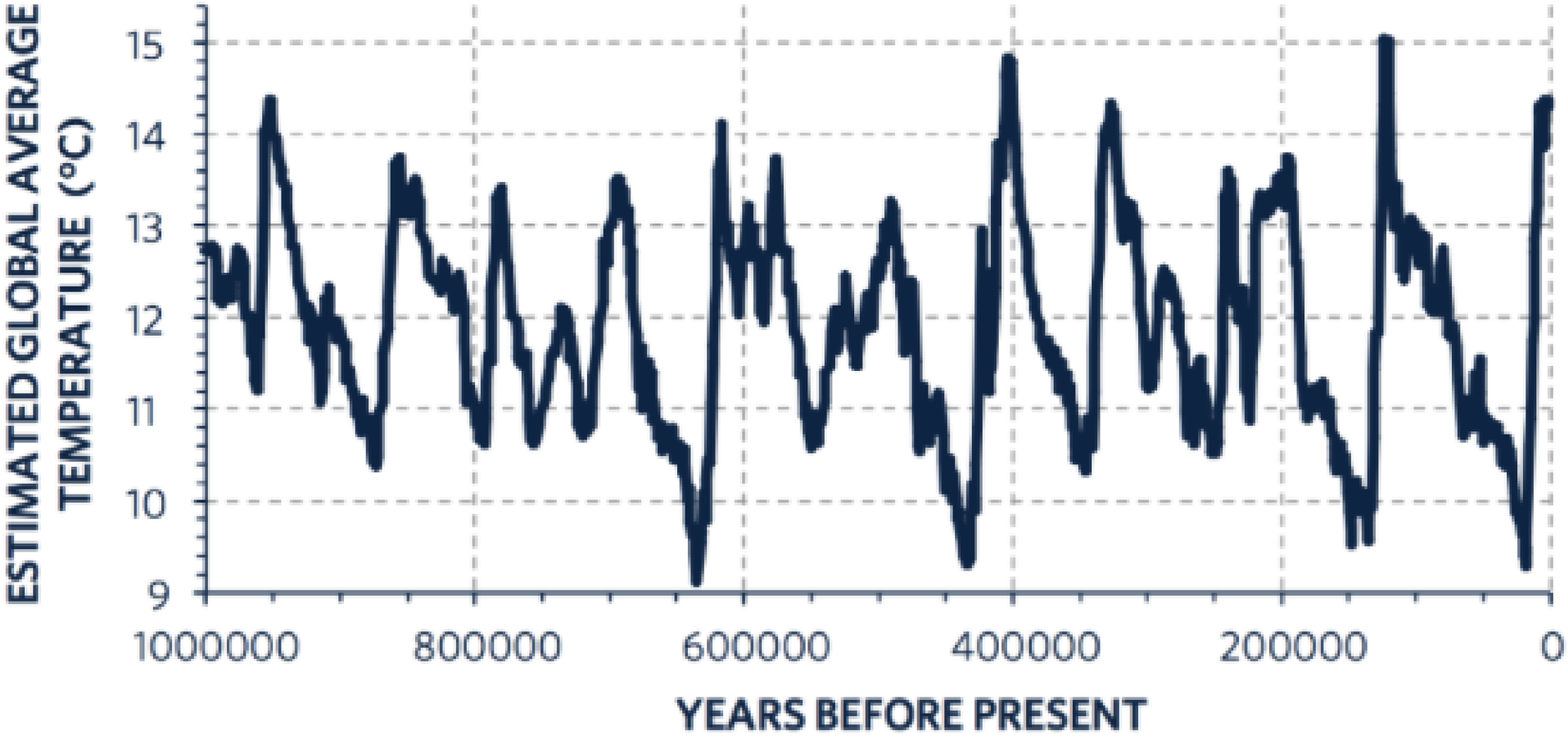
1. **Solar Variability**: Changes in the sun's energy output can influence Earth's climate. Solar cycles, which include variations in the sun's radiation and magnetic activity, can affect the amount of solar energy reaching the Earth.

2. **Orbital Variations**: Changes in Earth's orbit and axial tilt over long timescales, known as Milankovitch cycles, can affect the distribution and intensity of solar radiation received by the Earth, leading to changes in climate over tens of thousands to hundreds of thousands of years.

3. **Volcanic Activity**: Large volcanic eruptions can inject significant amounts of aerosol and sulphur dioxide into the atmosphere, leading to temporary cooling by blocking sunlight. These eruptions can cause short-term fluctuations in global temperatures.

4. **Ocean Circulation:** Ocean currents play a crucial role in redistributing heat around the Earth. Variations in ocean circulation patterns, such as El Niño and La Niña events in the Pacific Ocean, can have significant impacts on regional and global climate patterns.

Key Points of Natural Variability



Lecture Summary

- Earth's climate varies naturally over **short (years–decades)** and **long (thousands–100,000s of years)** timescales
- Natural variability arises from interactions within the **Sun–Earth system, atmosphere, oceans, cryosphere, and biosphere**
- Key drivers include **orbital variations (Milankovitch cycles), solar variability, volcanic activity, and ocean circulation**
- Milankovitch cycles (eccentricity, obliquity, precession) change the **distribution and seasonality of insolation**, driving ice ages
- **Ice cores and marine sediments** provide evidence of past temperature and greenhouse gas changes over hundreds of thousands of years
- Solar variability influences climate on **decadal timescales**, but shows no long-term increase in recent decades
- Volcanic eruptions cause **short-term global cooling** through sulphate aerosols in the stratosphere
- **ENSO** drives major interannual climate variability, influencing global weather patterns
- Natural variability cannot explain the **magnitude or rate of modern global warming**, which is dominated by anthropogenic forcing



Week 2 – What could I ask you in an exam?

1. Define natural variability within the context of the climate system.
2. Explain the drivers of climate variability, with examples.
3. Compare and contrast natural climate variability with anthropogenic climate change.
4. Summarise the key drivers of natural variability in the climate system and evaluate their relative importance.
5. Define short-term and long-term climate variability, and provide examples of each.
6. What evidence supports the occurrence of past ice ages in Earth's history?
7. Why is proxy data important for understanding past climate variability?
8. Describe how scientists use ice cores to reconstruct historical climate conditions.
9. What are Milankovitch cycles, and what impact do they have on Earth's long-term climate?
10. How does solar variability contribute to driving changes in the Earth's climate?
11. Explain how ocean circulation patterns, contribute to natural variability in the climate system.
12. What is the role of the El Niño-Southern Oscillation (ENSO) in shaping natural climate variability?
13. Discuss the concept of feedback mechanisms in the climate system, providing a detailed example.
14. How do aerosols from volcanic eruptions affect global temperatures, both in the short and long term?
15. Why is understanding natural climate variability essential for improving climate models and projections?

Global Climate Change

The University of Manchester

Week 1 – Introduction to climate change and its impacts (5th Feb)

Week 2 – Natural climate variability (12th Feb)

Week 3 – Radiation Law/Greenhouse Effect/Radiative forcing/feedbacks (19th Feb)

Week 4 – Anthropogenic impacts (26th Feb)

Week 5 – Observational evidence and uncertainties on climate change (5th March)

Week 6 – Mid Unit Test (10th March)

Week 7 – Climate Models (19th March)

Week 8 – Climate Change Adaptation (26th March)

Week 9 – Effects of Climate Change: Case Studies (23rd April)

Week 10 – Climate change Mitigation (30th April)

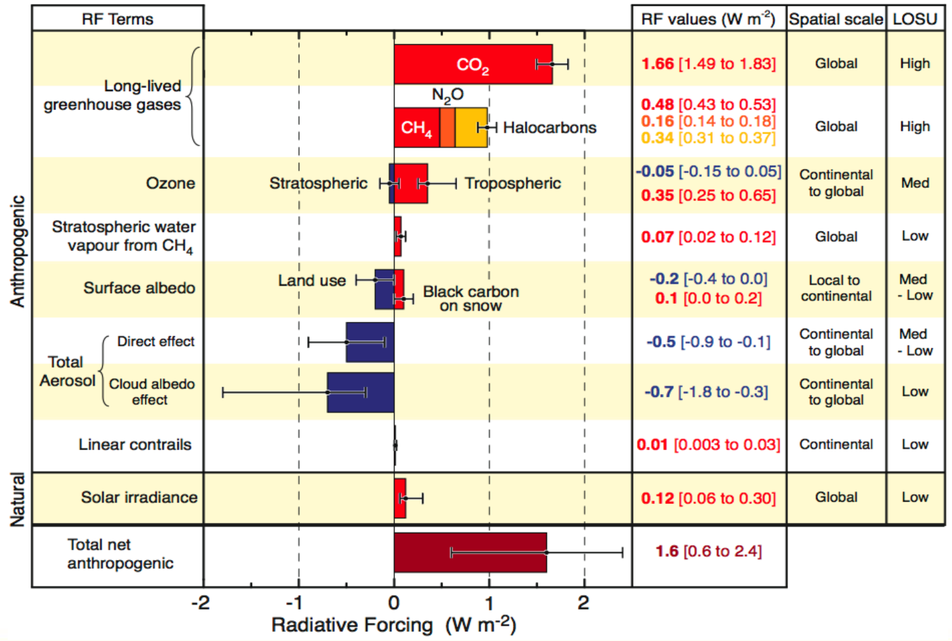
Week 11 – Revision Lecture (7th May)

Comments/Questions



MANCHESTER 1824
The University of Manchester

Next Week: Radiation, the Greenhouse Effect & Warming





Next Week: Radiation, the Greenhouse Effect & Warming

In next week's lectures, you will learn:

- How **blackbody radiation and energy balance** are used to calculate the equilibrium temperature of Earth and other planets.
- The physical basis of the **greenhouse effect**, including:
  - why the **Sun and Earth emit radiation at different wavelengths**, and
  - why **greenhouse gases** interact differently with incoming solar radiation and outgoing radiation.
- Why Earth's **actual surface temperature** is higher than its blackbody equilibrium temperature.
- The implications of **rising greenhouse gas concentrations** in the atmosphere.
- Why increasing greenhouse gases are expected to drive **further global warming**.

MANCHESTER 1824
The University of Manchester

Feedback

Join at menti.com | use code 5809 2619

MANCHESTER 1824

The University of Manchester

Thank you

Thomas.Bannan@Manchester.ac.uk

MANCHESTER 1824
The University of Manchester

Global Climate Change

Week 2, Questions and Answers

Dr Thomas Bannan

Q1.

What is the difference between natural variability and human-induced climate change? How do we separate their effects in climate studies?

Q2.

Which components of the Earth system interact to produce natural climate variability?

Q3.

How can volcanic eruptions serve as a "natural experiment" for studying climate response to aerosol forcing?

Q4.

How do scientists use ice cores to reconstruct long-term natural climate variability?

Q5.

MANCHESTER 1824
The University of Manchester

In what ways can understanding natural variability help us plan for extreme weather events?

Q6.

MANCHESTER 1824
The University of Manchester

Discuss the significance of Milankovitch cycles in shaping Earth's glacial and interglacial periods.

Q7.

MANCHESTER 1824
The University of Manchester

Can natural variability explain observed trends in global warming? Why or why not?

Q8.

How might future changes in natural variability interact with anthropogenic climate change to influence global climates?

Q9.

Compare the climatic influence of solar variability and volcanic eruptions. How do their impacts differ in duration and magnitude?

Q10.

The University of Manchester

Why can’t natural climate drivers explain the observed warming since the mid-20th century?